

SCOPE : Critical Review Q&A

108 Questions

v1 2026年9月 senkaL8

Introduction

This document is a critical Q&A designed to examine possible criticisms, counterexamples, alternative explanations, cases of non-identifiability, and risks of unfalsifiability concerning SCOPE.

Its purpose is not to prove that SCOPE is correct. Rather, it aims to distinguish between points that can be addressed by the current SCOPE framework and points that cannot yet be resolved. Competing explanations and conditions under which individual hypotheses would be weakened are stated where possible. Where such conditions cannot yet be established, no conclusion is forced; the issue is instead retained as “unresolved,” “not identifiable,” or “requiring further validation.” These are not omissions, but records of the framework’s current theoretical limitations.

This review is an internal audit conducted by the author. It does not constitute independent third-party validation or empirical confirmation of SCOPE’s validity. The fact that a phenomenon can be explained using SCOPE is not treated, by itself, as evidence that SCOPE is valid.

This document consists of 108 questions. The areas examined include SCOPE as a whole, individual layers, relationships between layers, the distinction between the roles of SCOPE and COS, functional correspondence between humans and LLMs, and the treatment of counterexamples and theoretical updates.

Q1. Is SCOPE ultimately a theory that can explain anything?

Related questions: Q2, Q3, Q4, Q5, Q94, Q98, Q103

The fact that SCOPE can organize, compare, or explain many phenomena does not itself demonstrate the validity of SCOPE or of its individual structural hypotheses. This distinction is already stated in the existing SCOPE paper.

However, to show that the current SCOPE is not an unfalsifiable framework simply because it can explain almost anything, it is necessary to specify the conditions under which SCOPE as a whole, or each of its structural hypotheses, would be weakened by observation, and to identify the observations that could distinguish SCOPE from competing models or alternative

structural hypotheses. At present, these conditions have not yet been sufficiently established.

The fact that a wide range of phenomena can be represented within SCOPE is not itself inconsistent with its design as a cognitive coordinate system. The problem arises if no observation can be identified that would count against SCOPE-specific structural claims, or if apparently contradictory observations can always be absorbed by changing interpretations or adding auxiliary explanations.

If such a situation persists, the structural hypotheses within SCOPE would lose strength as empirically testable hypotheses, and the problem of unfalsifiability would become more serious.

However, this would not necessarily mean that SCOPE as a whole immediately collapses. Even if some structural hypotheses are weakened, SCOPE may still retain value as a descriptive analytical framework for organizing and comparing cognitive phenomena within a common coordinate system.

Status:

“Partially addressed in the existing paper”

“Core unresolved issues remain”

“Falsification conditions need to be established”

Q2. If almost any phenomenon can be assigned to L0–L8, is there any value in describing it with SCOPE?

Related questions: Q1, Q3, Q4, Q5, Q13, Q94, Q103

The fact that a phenomenon can be described using SCOPE does not itself demonstrate the validity of SCOPE. This distinction is already stated in the existing paper.

If SCOPE has analytical value, that value does not come merely from assigning L0–L8 labels to phenomena. Rather, it lies in the possibility of using a common coordinate system to compare different cognitive structures and to distinguish and retain multiple competing hypotheses and unresolved components within the same analysis. The existing paper also explicitly describes an approach in which multiple hypotheses are retained when the same output could arise from different cognitive structures, rather than forcing an early single interpretation.

However, if SCOPE-based description amounts only to relabeling phenomena, and if it produces no additional distinctions, comparisons, or inferential possibilities beyond ordinary natural-language explanation or a simpler classification model, then the distinctive usefulness of SCOPE would be weakened.

If analyses using SCOPE consistently provide little or no additional information or discriminative value compared with simpler descriptive or classification methods, then the added value of using SCOPE as a cognitive coordinate system would be reduced.

Furthermore, if changing the layer configuration produces little meaningful change in the inferred cognitive structure or in the distinction between competing hypotheses, it becomes reasonable to question whether the coordinate placement itself carries meaningful analytical information.

However, this would not necessarily mean that SCOPE as a whole immediately collapses. It may still retain some value as a descriptive system for organizing cognitive phenomena, while the stronger claim that SCOPE provides unique or additional analytical value would be weakened.

Status:

“Partially addressed in the existing paper”

“Independent evaluation of usefulness is needed”

“Competing simpler classification models exist”

“Conditions for reduced usefulness can be specified”

Q3. Does explaining a phenomenon with SCOPE count as evidence that SCOPE is correct?

Related questions: Q1, Q2, Q4, Q5, Q94, Q103

In the current version of SCOPE, the ability to organize and compare many phenomena within a common framework is explicitly distinguished from the validity of SCOPE itself and of its individual structural hypotheses. Therefore, at least in the latest published text, SCOPE does not adopt the circular argument that “if SCOPE can explain something, SCOPE must be correct.”

However, making this distinction in the text and maintaining it in actual theoretical evaluation are separate issues.

Even if hallucinations, cognitive biases, or other phenomena can be structurally described using SCOPE, this shows only that those phenomena are not inconsistent with SCOPE. It does not directly demonstrate that SCOPE-specific structural hypotheses are supported or that SCOPE is superior to competing models.

Therefore, for explanatory examples to count as supporting evidence for SCOPE, more is required than the fact that an explanation can be constructed. Independent evaluation is needed, such as whether previously specified structural claims or discrimination criteria are consistent with observation, or whether SCOPE can distinguish differences that competing models cannot easily distinguish.

If the number of cases that can be explained using SCOPE continues to increase without comparison against contradictory observations or competing models, the accumulation of such cases provides limited support for the validity of SCOPE.

Furthermore, if explanations generated using SCOPE are then reused as evidence that SCOPE's structural hypotheses are correct, the theory would effectively be generating its own supporting evidence. This would create a circular structure and weaken the empirical support for SCOPE.

However, explanatory examples are not meaningless. They can still function as demonstrations of how the framework is used, how internally coherent an analysis is, and what range of phenomena can be described within it. The problem arises only when explanatory success is reinterpreted as evidence of validity.

Status:

"Principle already addressed in the existing paper"

"Continued auditing is required in actual validation"

"Risk of circular reasoning"

"External and comparative validation is needed"

Q4. Is SCOPE merely a post hoc explanation that assigns plausible layers after observing the output?

Related questions: Q1, Q2, Q3, Q5, Q7, Q8, Q9, Q13, Q14, Q94, Q98, Q99, Q103

SCOPE is not defined as a model that directly "discovers" internal cognitive structure from observable output. Instead, it uses observed output as evidence for provisionally placing multiple hypotheses across the layers and then evaluates those hypotheses in relation to the cognitive structure as a whole. The existing paper also explicitly states that SCOPE does not determine a single internal state from one observation and may retain multiple competing hypotheses.

Therefore, placing hypotheses after observing an output is not itself a flaw in SCOPE. It is part of the intended procedure for inferring internal structures that cannot be directly observed.

However, retrospective inference and an explanation that can be adjusted to fit any result are not the same thing.

SCOPE would become vulnerable to post hoc explanation if, after observing a result, an analyst selected only the most convenient layer configuration without considering competing configurations or hypotheses, and then changed not only the configuration but also the criteria used to select it whenever new observations appeared, so that the result could always be made consistent with the existing SCOPE framework.

In particular, if the criteria for selecting layer configurations, weakening or rejecting hypotheses, and comparing competing hypotheses remain unclear, the judgment that one interpretation has "higher coherence" may depend heavily on analyst discretion. In that case, SCOPE becomes increasingly difficult to distinguish from a post hoc explanatory system.

If there are no explicit criteria for determining which observations support or weaken competing hypotheses, and the preferred hypothesis becomes identifiable only after the outcome is known, it becomes difficult to distinguish SCOPE-based inference from an explanation selected to fit the observed result.

However, retrospective inference itself is not meaningless. Inferring possible causes from observed outcomes can have analytical value. The central issue for SCOPE is therefore not whether inference occurs after observation, but whether there are rules for comparing and rejecting hypotheses, and whether those rules themselves remain stable rather than being modified after the observation to preserve consistency with SCOPE.

Status:

“Inference procedure addressed in the existing paper”

“Criteria distinguishing inference from post hoc explanation remain underdeveloped”

“Risk of analyst dependence and unfalsifiability”

“Hypothesis selection and rejection criteria need clarification”

Q5. Under what conditions can SCOPE be judged to be wrong?

Related questions: Q1, Q3, Q4, Q84, Q85, Q94, Q100, Q102, Q103

In the current integrated SCOPE paper, SCOPE is positioned as an analytical framework for comparing and inferring the cognitive structures of humans and LLMs within the same coordinate system, using observable outputs as the common point of contact.

Therefore, the core defeat condition for SCOPE as a whole is not that each individual layer must be falsified one by one. Rather, it is that the central premise of treating humans and LLMs within the same cognitive coordinate system fails.

For example, if applying the same coordinate system to humans and LLMs requires the function assigned to the same layer to be defined differently for humans and for LLMs, then SCOPE’s core hypothesis of a common cognitive coordinate system is weakened.

If such incompatibility is not limited to a small number of layers or correspondences, and repeated examinations show that maintaining humans and LLMs within the same coordinate system is itself difficult, then SCOPE as a whole becomes subject to revision or rejection.

By contrast, if an individual layer, connection, or slide relationship is incompatible with observed results, then that specific proposition must be revised or rejected. Falsification of an individual proposition must therefore be distinguished from defeat of SCOPE as a whole.

Undefined concepts or new conditions must also not be introduced after a counterexample appears simply to preserve SCOPE. If this were allowed, SCOPE would effectively have no defeat condition.

At present, such a defeat condition for SCOPE as a whole is not explicitly defined in the integrated paper. In this Q&A, the failure of humans and LLMs to be compared and inferred

within the same cognitive coordinate system is treated as a core defeat condition for SCOPE as a whole.

Status:

“Core defeat condition established”

“Individual-proposition falsification distinguished from whole-framework defeat”

“Whole-framework defeat condition not explicitly defined in the current integrated paper”

“Post hoc avoidance of falsification is prohibited”

Q6. Is SCOPE a scientific theory, an analytical framework, or a cognitive coordinate system?

Related Qs: Q1, Q2, Q3, Q5, Q13, Q94, Q103

SCOPE is an analytical framework designed as a cognitive coordinate system.

“Cognitive coordinate system” describes SCOPE’s structural form. In SCOPE, L0–L8 are arranged as cognitive coordinates, while cognition is treated as an overall structure that also includes Malkuth and Daath. Observable outputs are used as clues for estimating the cognitive structure behind them.

“Analytical framework,” on the other hand, describes what this coordinate system is used for. SCOPE is used to estimate, compare, and analyze the cognitive structures of humans and LLMs within a shared coordinate system.

Therefore, “cognitive coordinate system” and “analytical framework” are not competing classifications. The former describes SCOPE’s structural form, while the latter describes its use and positioning.

At present, SCOPE as a whole is not positioned as an established scientific theory. However, this does not mean that SCOPE is outside the scope of validation.

SCOPE contains structural hypotheses that can be examined individually, including distinctions among layers, connection relationships, slide relationships, and functional correspondences between humans and LLMs.

Accordingly, in this Q&A, SCOPE is treated as an analytical framework designed as a cognitive coordinate system that contains structural hypotheses subject to validation.

Status:

“Cognitive coordinate system = SCOPE’s structural form”

“Analytical framework = SCOPE’s use and positioning”

“At present, SCOPE is not positioned as an established scientific theory”

“Its internal structural hypotheses are subject to validation”

Q7. If two people analyze the same output using SCOPE, will they arrive at the same inference?

Related Qs: Q8, Q9, Q10, Q11, Q13, Q14, Q103

Not necessarily.

SCOPE distinguishes between the observed output itself and the estimation of the cognitive structure behind it. Even when the observed output is the same, it may not be possible to determine a single hypothesis to place as the candidate explanation behind that output.

For example, the same statement, "I'm fine," may support multiple hypotheses about the cognitive state or structure behind it, such as "there is genuinely no problem," "the person is trying to reassure themselves," or "the person does not want to cause concern."

Therefore, the fact that two analysts propose different hypotheses from the same output does not in itself conflict with SCOPE's estimation method. However, this does not mean that all different hypotheses are equally valid.

When the available observations are insufficient to determine a single interpretation, SCOPE retains multiple hypotheses.

At the same time, because estimates may differ between analysts, the degree to which their estimation results agree requires separate evaluation.

Status:

"The same output does not guarantee the same estimation result"

"Observed output is distinguished from estimation of the cognitive structure behind it"

"Multiple hypotheses are retained when a single estimate cannot be determined"

"Different hypotheses are not necessarily equally valid"

"Inter-analyst reproducibility remains to be evaluated"

Q8. When different analysts propose different hypotheses, what criteria should be used to compare them?

Related Questions: Q7, Q9, Q10, Q11, Q12, Q13, Q103

In SCOPE, hypotheses are compared according to how well each of them is consistent with the observed information, the relationships among other layers, and the cognitive structure as a whole.

In SCOPE inference, hypotheses are provisionally placed at each layer on the basis of observed information, and their consistency with other layers and with the cognitive structure as a whole is examined. If a hypothesis is not sufficiently consistent, it may be revised or repositioned, and related hypotheses concerning other layers may also be reconsidered as necessary.

Here, revising or repositioning a hypothesis refers to an operation performed on an inferential hypothesis held by the analyst. This does not, by itself, mean that the cognitive structure of the person being analyzed has changed.

Therefore, when different analysts propose different hypotheses, each hypothesis is compared against the same observed information, the relationships among other layers, and the cognitive structure as a whole.

However, current SCOPE does not yet sufficiently define a specific evaluation method for decomposing “consistency” into distinct criteria or determining how much weight should be given to different kinds of inconsistency. This issue is addressed in Q9.

If the existing criteria do not allow multiple hypotheses to be distinguished, SCOPE does not force them into a single conclusion and instead retains multiple hypotheses.

Status:

“Different hypotheses are compared in terms of their consistency with observed information, relationships among other layers, and the cognitive structure as a whole.”

“The objects of revision or reconsideration are the inferential hypotheses held by the analyst. This does not, by itself, imply a change in the target’s cognitive structure.”

“A specific method for evaluating consistency has not yet been sufficiently defined.”

“When existing criteria cannot distinguish among multiple hypotheses, multiple hypotheses are retained.”

Q9. What Does It Specifically Mean for a Cognitive Structure to Have High Overall Consistency?

Related Qs: Q7, Q8, Q10, Q11, Q12, Q13, Q89, Q90, Q91, Q92, Q93, Q103

In SCOPE, “high consistency with the overall cognitive structure” means that a provisionally placed cognitive state is unlikely to conflict with the observed information, its relationships with other layers, and the cognitive structure as a whole.

SCOPE does not evaluate hypotheses for individual layers in isolation. Instead, it examines them as part of the combined cognitive structure. If placing a hypothesis causes consistency with connected layers or other hypotheses to break down, that hypothesis and related layers are reconsidered.

Therefore, the “most consistent cognitive structure” does not necessarily mean a structure that converges on a single correct answer. When the available observations do not allow hypotheses to be distinguished, multiple hypotheses may be retained.

At present, however, SCOPE does not sufficiently define a concrete evaluation method for measuring “consistency” on a common scale, such as how much weight should be given to different types of inconsistency.

For this reason, the expression “high consistency” should not be used as a justification for an analyst to select an arbitrary hypothesis. Rather, it should be treated as a perspective for comparing observed information and relationships among layers within the cognitive structure as a whole.

Status:

“Consistency = low conflict with observed information, relationships with other layers, and the overall cognitive structure”

“Layers are evaluated as part of the overall cognitive structure rather than in isolation”

“Highest consistency does not necessarily mean convergence on a single correct answer”

“When hypotheses cannot be distinguished, multiple hypotheses are retained”

“The concrete evaluation method for consistency is not yet sufficiently operationalized”

Q10. How Should Multiple Hypotheses Be Handled When They Are Equally Consistent?

Related Qs: Q7, Q8, Q9, Q11, Q12, Q13, Q103

When multiple hypotheses are similarly consistent and cannot be distinguished based on the currently available observations, SCOPE does not force a single conclusion and instead retains multiple hypotheses.

However, there are at least two different situations in which multiple possibilities may remain.

One is when the available observations are insufficient to distinguish among competing hypotheses.

The other is when multiple cognitive states are actually present at the same time.

SCOPE does not treat these two situations as identical. When the issue is insufficient information, the alternatives are retained as unresolved multiple hypotheses. When multiple states are estimated to be simultaneously present, they are retained as simultaneously present states.

These two situations are conceptually distinct, but in an individual case, the currently available observations may not be sufficient to determine which situation applies. In that case, SCOPE does not force a conclusion and leaves the issue unresolved.

Therefore, when multiple hypotheses are similarly consistent, SCOPE distinguishes them as far as the available observations allow and retains the remaining uncertainty as unresolved multiple hypotheses. When simultaneous presence of multiple states is estimated, those states are retained as simultaneously present.

Status:

“Similarly consistent hypotheses are not forced into a single conclusion”

“Uncertainty caused by insufficient information is distinguished from the simultaneous presence of multiple states”

“The distinction may itself remain unresolved in an individual case”

“When information is insufficient, multiple hypotheses are retained as unresolved”

“When multiple states are estimated to coexist, they are retained as simultaneously present states”

Q11. How far should inference go when observational information is insufficient?

Related Qs: Q7, Q8, Q9, Q10, Q12, Q13, Q14, Q103

In SCOPE, estimation should proceed only as far as the currently available observations allow the cognitive structure to be narrowed down, while anything beyond that remains unresolved.

SCOPE uses observed outputs to estimate which cognitive layers or structures may be involved behind them. It does not directly observe the internal states of the layers themselves.

Therefore, when the available observations do not allow a single hypothesis to be identified, multiple hypotheses are retained, and parts that cannot be determined from the observations are not forced into a definite state.

For example, the statement “I’m fine” alone may be consistent with multiple possible cognitive states behind it. In such a case, candidates that cannot be distinguished from the available observations are retained as unresolved multiple hypotheses rather than being forced into a single conclusion.

Accordingly, the scope of estimation in SCOPE is determined not by how much explanation can be constructed, but by how far the cognitive structure can be narrowed down from the currently available observations.

Status:

“Estimation proceeds only as far as the observations allow the possibilities to be narrowed down”

“Internal states are not treated as if they were directly observed”

“When a single hypothesis cannot be distinguished, multiple hypotheses are retained”

“Parts that cannot be determined remain unresolved”

“The scope of estimation is determined by how far the cognitive structure can be narrowed down from the available observations”

Q12. What is the basis for updating hypotheses when new observations are obtained?

Related Questions: Q7, Q8, Q9, Q10, Q11, Q13, Q90, Q91, Q92, Q103

Here, “updating a hypothesis” refers to revising or repositioning an inferential hypothesis that has been placed by the analyst. This does not, by itself, mean that the cognitive structure of the person being analyzed has changed. Likewise, the mere fact that a new observation has been obtained is not treated as evidence that the target’s cognitive structure has changed or that the analysis has moved from SCOPE to COS.

In SCOPE, a hypothesis is reconsidered not simply because a new observation exists, but on the basis of how well the existing hypothesis remains consistent with the observed information, the relationships among other layers, and the cognitive structure as a whole after the new observation is included in the evaluation.

In SCOPE inference, hypotheses are provisionally placed at each layer on the basis of observed information, and their consistency with other layers and with the cognitive structure as a whole is examined. If reevaluation including the new observation shows that the existing hypothesis is no longer sufficiently consistent, the analyst may reconsider that hypothesis and related hypotheses concerning other layers and reposition them as necessary.

Conversely, even when a new observation is consistent with an existing hypothesis, this alone does not establish that hypothesis as correct. If competing hypotheses are equally consistent, multiple hypotheses are retained.

Using multiple observations does not, by itself, require the description to be treated as COS. A description that uses multiple observations as evidence for comparing and inferring hypotheses about cognitive structure can still be treated within SCOPE.

By contrast, the ongoing process of updating a target model is treated as an area of connection between SCOPE and COS rather than as a role of SCOPE alone. However, the accumulation of multiple observations or the length of the observation period is not, by itself, a sufficient condition for treating the process as COS.

It is also not always possible to uniquely distinguish, from observable output alone, among a revision of the analyst’s inferential hypothesis, a change in the target’s cognitive structure, and a processing-level update within COS. At present, no prior criterion has been specified that can uniquely assign an observation to one of these categories. When they cannot be distinguished, the analysis does not force a conclusion in favor of any one of them.

Therefore, revising or repositioning hypotheses within SCOPE is not a matter of arbitrarily changing an explanation to fit new observations. It is based on reexamining the consistency between the observed information and the cognitive structure as a whole.

Status:

“The basis for reconsidering a hypothesis is not the mere existence of a new observation, but the consistency found after reevaluation.”

“Here, reconsideration refers to a revision of the inferential hypothesis held by the analyst. This does not, by itself, imply a change in the target’s cognitive structure.”

“Consistency with a new observation does not, by itself, establish a hypothesis.”

“When competing hypotheses are equally consistent, multiple hypotheses are retained.”

“Multiple observations or a longer observation period alone do not make a description COS.”
“The ongoing updating of a target model is treated as an area of connection between SCOPE and COS, but the accumulation of observations itself is not a sufficient condition.”
“No prior criterion has yet been specified for uniquely distinguishing among revision of the analyst’s inference, change in the target’s cognitive structure, and processing-level updating within COS. When they cannot be distinguished, no single interpretation is forced.”

Q13. Does SCOPE “discover” internal structures, or does it merely arrange competing hypotheses?

Related Qs: Q2, Q3, Q4, Q7, Q8, Q9, Q10, Q11, Q12, Q14, Q81, Q82, Q83, Q85, Q86, Q87, Q103

SCOPE is not a model that directly “discovers” unobservable internal structure.

SCOPE uses observable outputs, such as speech, text, behavior, and generated responses, to estimate which cognitive layers or structures may be involved behind them.

Therefore, layer configurations and cognitive structures estimated through SCOPE are not treated as observed facts themselves, but as hypotheses inferred from observed outputs.

At the same time, SCOPE does more than simply generate or list competing hypotheses. It uses the coordinate system to compare and reconsider them. Multiple hypotheses are provisionally placed and examined in relation to the observed information, relationships with other layers, and consistency with the cognitive structure as a whole.

Therefore, SCOPE’s estimation is not the direct discovery of internal structure, but the placement of hypotheses about internal structure based on observable outputs, followed by comparison and narrowing of those hypotheses according to the observed information, relationships with other layers, and consistency with the overall cognitive structure.

Even when a hypothesis is found to be the most consistent, it is not treated as if the internal structure itself had been directly observed.

Status:

“SCOPE is not a model that directly discovers internal structure”

“Observed facts are distinguished from hypotheses about internal structure”

“Competing hypotheses are not merely generated or listed, but compared and reconsidered”

“Hypotheses are compared and narrowed based on observed information, relationships with other layers, and consistency with the overall cognitive structure”

“Even the most consistent hypothesis is not treated as direct observation of internal structure”

Q14. To what extent can claims be made about each layer if the internal layers cannot be directly observed?

Related Qs: Q3, Q4, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q15–Q50, Q81–Q87, Q94, Q103

SCOPE does not claim that the internal layers from L0 to L8 can themselves be directly observed.

What can be directly observed are outputs such as speech, text, behavior, and generated responses. Hypotheses about the states of individual layers and the cognitive structure are estimated using these outputs as clues.

Therefore, statements such as “L4 made this judgment” or “L8 made this determination” cannot be treated as facts directly observed inside the system. More precisely, they must be treated as estimations, such as “this output is consistent with placing this hypothesis in L4” or “this cognitive structure is consistent with this hypothesis about L8.”

At the same time, how each layer is defined as a cognitive function within SCOPE can be described as part of the model's definition. However, the existence of such a definition is different from directly observing the state of that layer.

Therefore, what can currently be claimed about each layer is limited to how it is defined within SCOPE and which hypotheses about that layer can be placed based on the observed information, relationships with other layers, and consistency with the cognitive structure as a whole.

The layers themselves cannot be established as directly observed internal states.

Status:

“The internal layers themselves cannot be directly observed”

“Observed outputs are distinguished from estimations about internal layers”

“The definition of each layer can be described as part of the SCOPE model”

“Hypotheses about a layer can be placed based on observed information, relationships with other layers, and consistency with the overall cognitive structure”

“Each layer is not established as a directly observed internal state”

Q15. Why is L0 necessary as an independent coordinate?

Related Qs: Q14, Q16, Q17, Q18, Q81, Q82, Q83, Q85, Q86, Q87, Q94, Q103

In SCOPE, L0 is placed as an independent coordinate in order to distinguish cognition related to survival and life maintenance from other cognitive functions.

L0 deals with cognition related to survival and life maintenance, including hunger, the need for sleep, and danger avoidance. By placing these functions on an independent coordinate, they can be retained and compared as hypotheses distinct from other cognitive functions.

Status:

“L0 is placed to distinguish cognition related to survival and life maintenance from other cognitive functions.”

“By separating L0 as an independent coordinate, cognition related to survival and life maintenance can be retained and compared as a distinct hypothesis.”

Q16. Can L0 be replaced by other layers?

Related Qs: Q15, Q17, Q18, Q81, Q83, Q85, Q94, Q103

At present, it has not been sufficiently demonstrated that L0 cannot be replaced by other layers.

If other layers alone can distinguish and infer cognition related to survival and life maintenance to the same degree, then L0 can be regarded as replaceable.

Therefore, at present, it cannot be concluded that L0 is irreplaceable.

Status:

“It has not been sufficiently demonstrated that L0 is irreplaceable.”

“If other layers alone can distinguish and infer cognition related to survival and life maintenance to the same degree, L0 can be regarded as replaceable.”

“At present, it cannot be concluded that L0 is irreplaceable.”

Q17. What becomes indistinguishable if L0 is removed?

Related Qs: Q15, Q16, Q18, Q81, Q83, Q85, Q94, Q103

If L0 is removed, the distinction that allows cognition related to survival and life maintenance to be retained as a hypothesis separate from other cognitive functions is lost.

This does not mean that cognition related to survival and life maintenance can no longer be addressed at all. Without L0 as an independent coordinate, SCOPE can no longer distinguish these functions as occupying a separate position from other cognitive functions.

Status:

“If L0 is removed, the distinction that allows cognition related to survival and life maintenance to be retained as a separate hypothesis is lost.”

“Removing L0 does not mean that cognition related to survival and life maintenance can no longer be addressed.”

Q18. What observations would weaken the L0 hypothesis?

Related Qs: Q15, Q16, Q17, Q81, Q82, Q83, Q85, Q86, Q87, Q94, Q103

If cognition related to survival and life maintenance can be distinguished and inferred to the same degree without L0 as when L0 is included, the hypothesis that L0 should be treated as an independent coordinate would be weakened.

On the LLM side, responses that differ from ordinary responses are sometimes observed when the input directly concerns human life or physical safety. Based on these observations, a new candidate hypothesis is being considered: there may be an underlying functional role that could correspond to L0. However, it remains unresolved whether this behavior needs to be distinguished independently as L0, or whether it can be distinguished and inferred to the same degree using only the hypothesis of system coherence at L3.

If the L3 hypothesis alone is shown to handle the same observations to the same degree, what would first be weakened is the hypothesis that this behavior corresponds to L0 on the LLM side. This would not directly invalidate the independence of L0 on the human side.

Here, functional correspondence does not mean that humans and LLMs possess the same internal structure or the same existing module. It refers to a correspondence in functional roles that does not require identical implementations.

SCOPE is fundamentally designed to compare humans and LLMs within a shared cognitive coordinate system. Therefore, if sufficient investigation leads to the conclusion that no functional role corresponding to L0 can be established on the LLM side, the basic design of SCOPE as a shared coordinate system would need to be reconsidered.

Status:

“If the same level of distinction and inference is possible without L0 as with L0, the hypothesis of L0 as an independent coordinate would be weakened.”

“On the LLM side, unusual responses to inputs directly concerning human life or physical safety are being examined as evidence for a possible functional role corresponding to L0.”

“It remains unresolved whether this behavior requires an independent L0 distinction or can be handled to the same degree by the L3 hypothesis alone.”

“If the L3 hypothesis alone is sufficient, the LLM-side L0 correspondence hypothesis would be weakened first; this would not directly invalidate the independence of human L0.”

“Functional correspondence does not require identical internal structures, existing modules, or identical implementations.”

“If no functional role corresponding to L0 can be established on the LLM side after sufficient investigation, SCOPE’s basic design as a shared cognitive coordinate system would need to be reconsidered.”

Q19. Why is L1 necessary as an independent coordinate?

Related Qs: Q14, Q20, Q21, Q22, Q51, Q52, Q53, Q81, Q82, Q83, Q85, Q86, Q87, Q94, Q103

In SCOPE, L1 is placed as an independent coordinate in order to distinguish “verbalization” — the function that transforms cognitive content into a form that can be expressed linguistically or vocally — from other cognitive functions.

By treating L1 independently, SCOPE can separately organize and infer hypotheses about cognitive content and hypotheses about the state in which that content can be transformed into linguistic or vocal expression.

Status:

“L1 is placed in order to distinguish the function that transforms cognitive content into a linguistically or vocally expressible form from other cognitive functions.”

“Treating L1 independently makes it possible to separately organize and infer hypotheses about cognitive content and hypotheses about its expressible state.”

Q20. Can L1 be replaced by other layers?

Related Qs: Q19, Q21, Q22, Q51, Q52, Q53, Q81, Q83, Q85, Q94, Q103

At present, it has not been sufficiently demonstrated that L1 cannot be replaced by other layers.

In SCOPE, L1 is positioned as a coordinate that handles the function of transforming cognitive content into a form that can be expressed linguistically or vocally.

If it is shown that other layers alone can distinguish and infer, to the same degree, hypotheses about cognitive content and hypotheses about the state in which that content can be transformed into linguistic or vocal expression, then the verbalization function assigned to L1 can be regarded as replaceable by other layers.

Therefore, whether L1 is truly irreplaceable remains a subject for future verification.

Status:

“It has not yet been sufficiently demonstrated that L1 is irreplaceable.”

“If other layers alone can distinguish and infer hypotheses about cognitive content and hypotheses about its linguistically or vocally expressible state to the same degree, the verbalization function assigned to L1 can be regarded as replaceable.”

“The irreplaceability of L1 remains a subject for future verification.”

Q21. What distinctions would be lost if L1 were removed?

Related Qs: Q19, Q20, Q22, Q94, Q103

If L1 were removed, it would become more difficult to separately organize and infer hypotheses about cognitive content and hypotheses about the state in which that content can be transformed into linguistic or vocal expression.

In other words, treating L1 as an independent coordinate makes it possible to distinguish hypotheses about cognitive content from hypotheses about the function that transforms that content into a form that can be expressed linguistically or vocally.

Status:

“If L1 were removed, it would become more difficult to separately organize and infer hypotheses about cognitive content and hypotheses about its linguistically or vocally expressible state.”

“Treating L1 as an independent coordinate makes it possible to distinguish hypotheses about cognitive content from hypotheses about its transformation into linguistic or vocal expression.”

Q22. What observations would weaken the L1 hypothesis?

Related Qs: Q19, Q20, Q21, Q94, Q103

If it is shown that, even without separating L1, other layers alone can distinguish and infer, to the same degree, hypotheses about cognitive content and hypotheses about the state in which that content can be transformed into linguistic or vocal expression, the hypothesis that L1 should be treated as an independent coordinate would be weakened.

Likewise, if adding L1 produces no substantial difference in the states that can be distinguished with respect to verbalization or in the resulting inferences, the basis for treating L1 as an independent coordinate would also be weakened.

Status:

“If the same distinctions can be organized and inferred to the same degree using other layers without separating L1, the hypothesis that L1 should be treated as an independent coordinate is weakened.”

“If adding L1 produces no substantial difference in the states distinguishable with respect to verbalization or in the resulting inferences, the basis for treating it as an independent coordinate is weakened.”

Q23. Why is L2 necessary as an independent coordinate?

Related Qs: Q24, Q25, Q26, Q51, Q52, Q53, Q94, Q103

In SCOPE, L2 is defined as an independent coordinate in order to organize and estimate emotional configuration toward a target separately from other cognitive functions.

For example, even when the same judgment or theory is held regarding the same target, the emotional configuration toward that target may differ, such as liking, anxiety, or anger.

By treating L2 independently, these differences in emotional configuration can be organized and estimated as structural hypotheses separate from other cognitive functions.

Status:

“L2 is positioned to distinguish emotional configuration toward a target from other cognitive functions.”

“L2 is not a coordinate that generates emotion itself.”

“Even when the same judgment or theory is held, different emotional configurations can be treated as separate structural hypotheses.”

Q24. Can L2 be replaced by other layers?

Related Qs: Q23, Q25, Q26, Q51, Q52, Q53, Q94, Q103

At present, it has not been sufficiently demonstrated that L2 cannot be replaced by other layers.

In SCOPE, L2 is positioned as a coordinate for organizing and estimating emotional configuration toward a target. However, if a structure without L2 as an independent coordinate could use other layers to organize and estimate emotional configuration as a structural hypothesis separate from other cognitive functions to the same degree, the necessity of maintaining L2 as an independent coordinate would be weakened.

Therefore, whether L2 cannot be replaced by other layers remains a subject for future comparison and validation.

Status:

“It has not yet been sufficiently demonstrated that L2 cannot be replaced by other layers.”

“If a structure without L2 as an independent coordinate can use other layers to organize and estimate emotional configuration to the same degree, the non-substitutability of L2 would receive less support.”

“The non-substitutability of L2 remains a subject for future comparison and validation.”

Q25. What distinctions would be lost if L2 were removed?

Related Qs: Q23, Q24, Q26, Q51, Q52, Q53, Q81, Q83, Q85, Q94, Q103

If L2 were removed, SCOPE would lose an independent coordinate for treating emotional configuration toward a target as a structural hypothesis separate from other cognitive functions.

For example, even when the same judgment or theory is held regarding the same target, the emotional configuration toward that target may differ, such as liking, anxiety, or anger. L2 is a coordinate for organizing and estimating these differences separately from other cognitive functions.

LLMs are not assumed to have the same emotions as humans. However, in interpersonal responses, response tendencies may change according to the other person’s emotional state as inferred from their statements and context. SCOPE treats these changes in response tendencies as a functional correspondence with L2.

Therefore, if L2 were removed, SCOPE would lose an independent coordinate for separately representing human emotional configuration toward a target and changes in LLM response tendencies that are hypothetically treated as functionally corresponding to L2.

Status:

“Removing L2 would eliminate the independent coordinate used to treat emotional configuration toward a target as a structural hypothesis.”

“Even when the same judgment or theory is held, differences in emotional configuration toward the target can be treated separately.”

“In LLMs, response tendencies may change according to the other person’s emotional state as inferred from statements and context.”

“SCOPE hypothetically treats these changes in LLM response tendencies as a functional correspondence with L2.”

Q26. What observations would weaken the L2 hypothesis?

Related Qs: Q23, Q24, Q25, Q51, Q52, Q53, Q81, Q83, Q85, Q94, Q103

The hypothesis that L2 is necessary as an independent coordinate would be weakened if observations consistently showed no substantial difference in the phenomena that can be distinguished or in the resulting estimates when L2 is used versus when it is not.

For example, if differences in emotional configuration, such as liking, anxiety, or anger, do not yield any additional distinguishable phenomena or different estimation results when L2 is used as an independent coordinate compared with a structure that does not include L2 as an independent coordinate, the basis for treating L2 independently would be weakened.

For LLMs, if changes in response tendencies that are hypothetically treated as functionally corresponding to L2 can be organized and estimated equally well without positing that correspondence, the functional correspondence hypothesis would also be weakened.

Therefore, a major condition that would weaken the hypothesis of L2’s independence is that using L2 as an independent coordinate fails to produce additional distinguishable phenomena or estimation results through observation.

Status:

“If observations consistently show no substantial difference in distinguishable phenomena or estimation results between using and not using L2, the hypothesis of L2’s independence would be weakened.”

“If adding L2 does not produce additional distinguishable phenomena or estimation results, the basis for treating it as an independent coordinate would be weakened.”

“If no substantial difference is found on the LLM side with or without the proposed correspondence to L2, the functional correspondence hypothesis would be weakened.”

“If treating L2 as an independent coordinate does not yield additional distinguishable phenomena or estimation results, this would be a major condition weakening the hypothesis of its independence.”

Q27. Why is L3 necessary as an independent coordinate?

Related Qs: Q28, Q29, Q30, Q67, Q68, Q69, Q70, Q71, Q72, Q73, Q94, Q103

In SCOPE, L3 is positioned as an independent cognitive coordinate that handles self-consistency within the individual.

L3 handles the function of organizing and maintaining values, beliefs, worldview, self-image, norms, and related elements in a form with as little internal contradiction as possible. This is distinguished from other cognitive functions such as emotional configuration, boundary setting, acceptance, and theory construction.

In SCOPE, this kind of self-consistency involving values, beliefs, worldview, and self-image is placed in L3 separately from other cognitive functions.

For LLMs, SCOPE does not assume an internal structure identical to human self-consistency. Instead, the function of maintaining consistency across responses and preserving basic system-level policies is treated as a functional correspondence with L3.

However, it has not yet been sufficiently verified that this form of self-consistency must be represented by an independent L3 coordinate. If the same differences can be organized and inferred to the same degree using other layers or combinations of those layers, the necessity of treating L3 as independent would be weakened.

Therefore, at present, the design rationale for separating L3 is to treat human self-consistency involving values, beliefs, worldview, and self-image, and, for LLMs, consistency across responses and basic system-level policies, as cognitive axes distinct from other cognitive functions.

Status:

“L3 is positioned to handle self-consistency involving values, beliefs, worldview, self-image, norms, and related elements.”

“L3 is distinguished from cognitive functions such as emotional configuration, boundary setting, acceptance, and theory construction.”

“For LLMs, consistency across responses and basic system-level policies is treated as a functional correspondence.”

“Defining L3 as independent and verifying that its independence is necessary are treated as separate issues.”

“If the same differences can be handled using other layers or combinations of them, the necessity of treating L3 as independent would be weakened.”

Q28. Can L3 be replaced by other layers?

Related Qs: Q27, Q29, Q30, Q67, Q68, Q69, Q70, Q72, Q73, Q94, Q103

At present, it has not been sufficiently verified that L3 cannot be fully replaced by other layers.

In SCOPE, self-consistency—such as organizing and maintaining values, beliefs, worldview, self-image, norms, and related elements in a form with as little internal contradiction as possible—is treated independently as L3.

However, it remains possible that a combination of the states and interactions of other layers could organize and infer an equivalent self-consistency function to the same degree. If so, the necessity of treating L3 as an independent coordinate would be weakened.

For LLMs, the same applies. If consistency across responses and basic system-level policies could be handled to the same degree through combinations of other cognitive functions alone, the need to posit an independent functional correspondence with L3 would be weakened.

At present, the range of observations is limited, and comparative testing between a structure that includes L3 and an alternative model composed only of combinations of other layers is still insufficient.

Therefore, although L3 is currently treated as an independent coordinate, it has not yet been established that it cannot be replaced by combinations of other layers.

Status:

“It has not yet been sufficiently verified that L3 cannot be fully replaced by other layers.”

“In SCOPE, self-consistency is treated independently as L3.”

“It remains possible that combinations of other layers could handle an equivalent self-consistency function to the same degree.”

“If the same function can be handled on the LLM side through combinations of other cognitive functions, the necessity of an independent functional correspondence with L3 would be weakened.”

“At present, the range of observations is limited, and comparison between models with and without L3 remains insufficient.”

“Whether L3 is non-replaceable remains an open question for future verification.”

Q29. What distinctions would be lost if L3 were removed?

Related Qs: Q27, Q28, Q30, Q67, Q68, Q69, Q70, Q72, Q73, Q94, Q103

If L3 were removed, one distinction that could be lost is the ability to treat self-consistency—such as organizing and maintaining values, beliefs, worldview, self-image, norms, and related elements in a form with as little internal contradiction as possible—as separate from other cognitive functions.

Without L3, it may become more difficult to place differences in the self-consistency formed within an individual, including values, beliefs, worldview, and self-image, as an independent structural hypothesis.

For LLMs, what could be lost is the distinction that treats consistency across responses and basic system-level policies separately from other cognitive functions, rather than treating them as equivalent to human self-consistency.

However, it has not yet been sufficiently verified that this distinction can only be preserved through L3.

Therefore, the main distinction that could be lost by removing L3 is the ability to treat human self-consistency, and for LLMs, consistency across responses and the maintenance of basic system-level policies, separately from other cognitive functions.

Status:

“Removing L3 could make it more difficult to treat self-consistency as an independent structural hypothesis.”

“Self-consistency involving values, beliefs, worldview, self-image, norms, and related elements can be treated separately from other cognitive functions.”

“For LLMs, consistency across responses and basic system-level policies are treated as a functional correspondence.”

“Removing L3 could make it more difficult to treat this consistency-maintenance function independently on the LLM side.”

“It has not yet been sufficiently verified that this distinction can only be preserved through L3.”

Q30. What observations would weaken the L3 hypothesis?

Related Qs: Q27, Q28, Q29, Q67, Q68, Q69, Q70, Q71, Q72, Q73, Q94, Q103

The L3 hypothesis would be weakened if the same observable differences could be organized and inferred to the same degree without treating self-consistency as an independent cognitive coordinate.

For example, if self-consistency involving values, beliefs, worldview, self-image, norms, and related elements could be sufficiently handled through combinations of other layers alone, and adding L3 produced no difference in distinction or inference, the necessity of treating L3 as independent would be weakened.

For LLMs, the same applies. If consistency across responses and basic system-level policies could be handled to the same degree through other cognitive functions without positing a correspondence with L3, the basis for treating these functions as functionally corresponding to L3 would be weakened.

If a model that includes L3 and a model that does not include L3 show no practical difference in how observable results are organized or inferred, this would also count as an observation that weakens the L3 hypothesis.

Therefore, a primary defeat condition for the L3 hypothesis is that removing L3 would not result in the practical loss of any additional distinction gained by treating self-consistency independently.

Status:

“L3’s independence would be weakened if the same observable differences could be organized and inferred to the same degree without L3.”

“If self-consistency could be sufficiently handled through combinations of other layers alone, the necessity of treating L3 as independent would be weakened.”

“If the same functions could be handled on the LLM side without positing a correspondence with L3, the basis for the functional correspondence would be weakened.”

“If models with and without L3 show no practical difference in organizing or inferring observable results, the L3 hypothesis would be weakened.”

“A primary defeat condition is that removing L3 would not result in the practical loss of any additional distinction gained by treating self-consistency independently.”

Q31. Why Is L4 Necessary as an Independent Coordinate?

Related Qs: Q32, Q33, Q34, Q54, Q55, Q56, Q57, Q61, Q63, Q64, Q94, Q103

In SCOPE, L4 is positioned as an independent coordinate for handling boundary setting with cognitive objects.

L4 is necessary because phenomena that appear different on the surface, such as checking information, keeping distance from another person, and viewing oneself objectively, can be organized under the common cognitive function of how boundaries are set with cognitive objects. This function can then be distinguished from other cognitive functions.

For LLMs as well, behaviors such as not simply accepting input as given and instead checking it, or producing one response from multiple candidates, are treated as functionally corresponding to L4.

Therefore, L4 is positioned as a coordinate for organizing and inferring boundary setting with cognitive objects separately from other cognitive functions.

Status:

“L4 is positioned to handle boundary setting with cognitive objects as an independent function.”

“Information checking, interpersonal distance, and objective self-observation are organized under the common function of boundary setting.”

“In LLMs, behaviors such as checking input rather than simply accepting it, or producing one response from multiple candidates, are treated as functionally corresponding to L4.”

Q32. Can L4 Be Replaced by Other Layers?

Related Qs: Q31, Q33, Q34, Q54, Q55, Q56, Q57, Q61, Q63, Q64, Q94, Q103

At present, it has not been shown that other layers alone can handle the same boundary-setting function as L4.

L4 handles boundary setting with cognitive objects. In SCOPE, phenomena such as checking information, keeping distance from another person, and viewing oneself objectively are organized under this common function.

Information processed by other layers may serve as material for L4’s judgment, but this does not mean that those other layers themselves perform the boundary-setting function.

However, if a model without L4 can organize and infer observable differences related to boundary setting to the same degree as a model that includes L4, the need to treat L4 as an independent coordinate would be weakened.

Therefore, whether L4 can be replaced remains unconfirmed, and whether the same function can be handled by other layers alone remains a subject for comparative testing.

Status:

“It has not yet been shown that other layers alone can handle the same boundary-setting function as L4.”

“Other layers providing information for L4’s judgment is distinct from replacing the boundary-setting function itself.”

“If a model without L4 can organize and infer observable differences as well as a model with L4, the independence of L4 would be weakened.”

“The replaceability of L4 remains a subject for comparative testing.”

Q33. What Becomes Difficult to Distinguish If L4 Is Removed?

Related Qs: Q31, Q32, Q34, Q54, Q55, Q56, Q57, Q61, Q63, Q64, Q94, Q103

If L4 is removed, it becomes difficult to organize and infer boundary setting with cognitive objects as a cognitive function distinct from other cognitive functions.

In SCOPE, phenomena that appear different on the surface, such as checking information, keeping distance from another person, and viewing oneself objectively, are organized under L4 as expressions of the common function of boundary setting with cognitive objects.

Without L4, the coordinate for organizing these phenomena separately from other cognitive functions would be lost, making it more difficult to treat differences arising from boundary setting itself as an independent distinction.

For LLMs as well, it would become more difficult to distinguish, as an independent functional correspondence, behaviors such as checking input rather than simply accepting it, or producing one response from multiple candidates.

Therefore, what is lost by removing L4 is the coordinate for distinguishing boundary setting with cognitive objects as an independent cognitive function.

Status:

“If L4 is removed, boundary setting itself becomes more difficult to organize and infer separately from other cognitive functions.”

“Information checking, interpersonal distance, and objective self-observation become more difficult to organize under the common function of boundary setting.”

“In LLMs, behaviors such as checking input rather than simply accepting it, or producing one response from multiple candidates, become more difficult to distinguish as an independent functional correspondence.”

“What is lost is the coordinate for treating boundary setting with cognitive objects as an independent cognitive function.”

Q34. What Observations Would Weaken the L4 Hypothesis?

Related Qs: Q31, Q32, Q33, Q54, Q55, Q56, Q57, Q61, Q63, Q64, Q94, Q103

The L4 hypothesis would be weakened if the same observable differences could be organized and inferred to a similar degree without treating boundary setting as an independent cognitive coordinate.

For example, if differences involving information verification, interpersonal distance, and objective self-observation could be organized and inferred to a similar degree using other layers alone, or if the same observable distinctions could be maintained even when L4 and L5 were treated as a single function, the need to maintain L4 as an independent coordinate would be weakened.

For LLMs as well, if behaviors such as checking input information or selecting one response from multiple candidates could be organized and inferred to a similar degree without positing a functional correspondence with L4, the basis for that correspondence hypothesis would be weakened.

Therefore, a major defeat condition for the hypothesis that L4 should be maintained as an independent coordinate would be the absence of any additional observable distinction gained by separating L4. However, showing that L4 and L5 do not need to be separated would not by itself show that a boundary-setting function does not exist.

Status:

“If the same observable differences can be organized and inferred to a similar degree using other layers alone, the independence of L4 is weakened.”

“If the same observable distinctions can be maintained without separating L4 and L5, the need to treat them as independent coordinates is weakened.”

“If the same LLM behaviors can be organized and inferred to a similar degree without positing correspondence with L4, the basis for the functional correspondence hypothesis is weakened.”

“The absence of additional distinctions gained by separating L4 is a major defeat condition for the L4 independence hypothesis.”

“The claim that L4 and L5 need not be separated must be distinguished from the claim that a boundary-setting function itself does not exist.”

Q35. Why Is L5 Necessary as an Independent Coordinate?

Related Qs: Q33, Q34, Q36, Q37, Q38, Q54, Q55, Q56, Q57, Q81, Q85, Q94, Q103

L5 is treated as an independent coordinate in order to distinguish and infer differences in how an object is accepted into cognition from other cognitive functions.

In SCOPE, “acceptance” in L5 does not mean liking an object, judging it to be true or good, or agreeing with it. It refers to the process by which people, information, values, theories, the self, and other objects are incorporated into cognition.

For example, the same advice may be accepted differently depending on who provides it. L5 is used to organize such differences in acceptance that cannot be explained by content alone.

L5 is not a directly observable internal mechanism. It is a hypothetical coordinate inferred from observable outputs such as speech and behavior.

Status:

“L5 is a coordinate for organizing and inferring differences in how objects are accepted into cognition.”

“Acceptance in L5 does not itself mean liking, judging something as true or good, or agreeing with it.”

“L5 addresses observable differences in how the same content may be accepted depending on who provides it.”

“L5 is not directly observed, but is hypothetically inferred from observable output.”

Q36. Can L5 Be Replaced by Other Layers?

Related Qs: Q31, Q32, Q35, Q37, Q38, Q54, Q55, Q56, Q57, Q63, Q64, Q81, Q85, Q94, Q103

In the current SCOPE model, L5 is treated as an independent coordinate for the function of accepting an object into cognition.

Other layers may influence acceptance, but influencing acceptance is not the same as being able to describe acceptance itself equally well through another layer or a combination of layers.

L5 does not refer to liking an object or judging whether it is true, false, good, or bad. It refers to the function by which people, information, values, theories, the self, and other objects are taken into cognition.

However, it has not been demonstrated that L5 cannot be replaced by other layers. If another layer or a combination of layers can preserve the same observable distinctions to a similar degree without using L5, the need to maintain L5 as an independent coordinate would be weakened.

This would not mean that a function corresponding to acceptance does not exist. What would be weakened is the need to separate that function as an independent coordinate called L5.

Therefore, SCOPE currently treats L5 as an independent coordinate for acceptance, while the possibility of replacement by other layers remains open to comparative testing.

Status:

“In the current SCOPE model, L5 is treated as an independent coordinate for acceptance.”

“Other layers influencing acceptance is different from replacing acceptance itself.”

“Replacement by combinations of multiple layers, not only by a single layer, is also a valid comparison.”

“If the same observable distinctions can be preserved to a similar degree without L5, the need to maintain L5 as an independent coordinate is weakened.”

“Weakening the independence of L5 is different from denying the existence of a function corresponding to acceptance.”

Q37. What Would No Longer Be Distinguishable If L5 Were Removed?

Related Qs: Q33, Q35, Q36, Q38, Q54, Q55, Q56, Q57, Q61, Q63, Q64, Q81, Q85, Q94, Q103

If L5 is removed, it becomes harder to organize and infer differences in how an object is accepted into cognition separately from other cognitive functions.

In SCOPE, L5 refers to the function by which people, information, values, theories, the self, and other objects are taken into cognition. This acceptance is distinguished from liking, judgments of truth or goodness, and agreement itself.

Without L5, observable differences such as the same information being accepted differently would be harder to organize from the independent perspective of “acceptance,” and the distinction between acceptance and emotion, judgment, or boundary setting would become weaker.

For LLMs as well, it would become harder to separately organize behaviors concerning how input information or premises are taken in as objects for subsequent processing as a possible functional correspondence with L5.

Therefore, what is lost by removing L5 is an independent coordinate for distinguishing differences in how objects are accepted into cognition.

Status:

“If L5 is removed, observable differences related to acceptance become harder to organize and infer separately from other cognitive functions.”

“The coordinate that distinguishes acceptance from liking, judgments of truth or goodness, and agreement would be lost.”

“Observable differences in how the same information is accepted would become harder to organize independently as acceptance.”

“For LLMs, behaviors concerning how input information or premises are taken in for subsequent processing would become harder to organize independently as a possible functional correspondence with L5.”

“What is lost is an independent coordinate for distinguishing differences in how objects are accepted into cognition.”

Q38. What Observations Would Weaken the L5 Hypothesis?

Related Qs: Q34, Q35, Q36, Q37, Q54, Q55, Q56, Q57, Q63, Q64, Q81, Q85, Q94, Q98, Q99, Q100, Q102, Q103

The hypothesis that L5 should be treated as an independent coordinate would be weakened if differences related to acceptance could be organized and inferred to a similar degree without using L5.

For example, if differences in how people, information, values, theories, the self, and other objects are taken into cognition could be distinguished to a similar degree using other cognitive functions such as emotion, judgments of truth or goodness, or boundary setting, and no additional distinction gained from L5 could be identified, the independence of L5 would be weakened.

If L4 and L5 could also be treated as a single function while preserving the same observable distinctions as when they are separated, the need to treat them as separate coordinates would be weakened. However, this would only show that the separation between L4 and L5 may be unnecessary; it would not show that only one of them is unnecessary.

For LLMs as well, if behaviors involving how input information or premises are taken in for subsequent processing could be organized to a similar degree without positing a functional correspondence with L5, that correspondence hypothesis would be weakened.

However, none of these observations would by themselves mean that a function corresponding to acceptance does not exist. What would be weakened is the independence of L5, the need to separate L4 and L5, or the hypothesis of functional correspondence with LLMs.

Status:

“If the same observable distinctions can be maintained without L5, the hypothesis that L5 should be treated as an independent coordinate is weakened.”

“If other cognitive functions can preserve the same distinctions and no additional distinction gained from L5 can be identified, the independence of L5 is weakened.”

“If L4 and L5 can be treated as a single function while preserving the same distinctions, the need to separate them as independent coordinates is weakened.”

“In that case, it cannot be concluded that only one of the two is unnecessary.”

“If LLM behavior can be organized to a similar degree without correspondence with L5, the functional correspondence hypothesis is weakened.”

“Weakening the independence of L5 is different from denying the existence of a function corresponding to acceptance.”

Q39. Why Is L6 Necessary as an Independent Coordinate?

Related Qs: Q40, Q41, Q42, Q47, Q48, Q58, Q59, Q60, Q61, Q63, Q64, Q81, Q85, Q94, Q103

L6 is treated as an independent coordinate in order to distinguish and infer differences in what kinds of explanatory models or theories are constructed about a cognitive object.

In SCOPE, “theory construction” in L6 does not refer only to academic theories. It includes everyday reasoning, causal explanations, comparisons, predictions, structuring, and other cognitive models constructed to explain an object.

For example, different people may construct different reasons or explanations for the same event. L6 is a coordinate for organizing such differences in explanatory models that may not be fully captured by differences in knowledge alone.

L6 is not a coordinate that determines truth or falsehood, nor does it guarantee that a constructed explanatory model corresponds to reality or is logically correct. L6 is also not a directly observable internal mechanism, but is hypothetically inferred from observable output.

Status:

“L6 is a coordinate for organizing and inferring differences in explanatory models or theories constructed about an object.”

“L6 theory construction includes not only academic theories, but also reasoning, causal explanation, comparison, prediction, and structuring.”

“L6 addresses observable differences in the explanations constructed for the same event.”

“Such differences may not be fully captured by differences in knowledge alone.”

“L6 does not determine truth or guarantee that explanatory models correspond to reality or are logically correct.”

“L6 is not directly observed, but is hypothetically inferred from observable output.”

Q40. Can L6 Be Replaced by Other Layers?

Related Qs: Q39, Q41, Q42, Q47, Q48, Q58, Q59, Q60, Q63, Q64, Q81, Q85, Q94, Q103

In the current SCOPE model, L6 is treated as an independent coordinate for the function of constructing explanatory models or theories about a cognitive object.

Other layers may influence theory construction, but this is different from being able to preserve the same observable distinctions using another layer or a combination of layers without L6.

However, it has not been demonstrated that L6 cannot be replaced by other layers. If another layer or a combination of layers can preserve the same observable distinctions to a similar degree without L6, the need to maintain L6 as an independent coordinate would be weakened.

This would not mean that the function of constructing explanatory models or theories does not exist. What would be weakened is the need to separate that function as an independent coordinate called L6.

Therefore, SCOPE currently treats L6 as an independent coordinate, while the possibility of replacement by other layers remains open to comparative testing.

Status:

“In the current SCOPE model, L6 is treated as an independent coordinate for constructing explanatory models or theories.”

“Other layers influencing L6 is different from preserving the same observable distinctions without L6.”

“Replacement by combinations of multiple layers, not only by a single layer, is also a valid comparison.”

“If the same observable distinctions can be preserved without L6, the need to maintain L6 as an independent coordinate is weakened.”

“Weakening the independence of L6 is different from denying the existence of a function that constructs explanatory models or theories.”

Q41. What Would No Longer Be Distinguishable If L6 Were Removed?

Related Qs: Q39, Q40, Q42, Q47, Q48, Q58, Q59, Q60, Q61, Q63, Q64, Q81, Q85, Q94, Q103

If L6 is removed, it becomes harder to organize and infer differences in what kinds of explanatory models or theories are constructed about a cognitive object separately from other cognitive functions.

For example, even when different explanations are given for the same event, there would no longer be an independent coordinate for organizing them in terms of what explanatory model has been constructed. As a result, differences appearing in reasoning, causal explanation, comparison, prediction, and structuring would become harder to distinguish from differences related to other cognitive functions.

For LLMs as well, behaviors in which explanations are constructed from input information or premises would become harder to organize independently as a possible functional correspondence with human L6.

Therefore, what is lost by removing L6 is an independent coordinate for distinguishing differences in explanatory models or theories constructed about a cognitive object.

Status:

“If L6 is removed, observable differences related to explanatory models or theories become harder to organize and infer separately from other cognitive functions.”

“The coordinate for independently treating reasoning, causal explanation, comparison, prediction, and structuring as differences in explanatory models would be lost.”

“Observable differences in explanations given for the same event would become harder to organize and infer independently as hypotheses about L6.”

“For LLMs, behaviors in which explanations are constructed would become harder to organize independently as a possible functional correspondence with human L6.”

“What is lost is an independent coordinate for distinguishing differences in explanatory models or theories constructed about a cognitive object.”

Q42. What Observations Would Weaken the L6 Hypothesis?

Related Qs: Q39, Q40, Q41, Q47, Q48, Q58, Q59, Q60, Q63, Q64, Q81, Q85, Q94, Q98, Q99, Q100, Q102, Q103

The hypothesis that L6 should be treated as an independent coordinate would be weakened if differences related to explanatory models or theories could be organized and inferred to a similar degree without using L6.

For example, if observable differences in the reasons or explanations given for the same event could be distinguished to a similar degree using other cognitive functions or combinations of them, and adding L6 produced no additional observable distinction, the need to maintain L6 as an independent coordinate would be weakened.

For LLMs as well, if behaviors in which explanations are constructed from input information or premises could be organized to a similar degree without positing a functional correspondence with human L6, that correspondence hypothesis would be weakened.

However, this would not mean that a function corresponding to the construction of explanatory models or theories does not exist. What would be weakened is the hypothesis that this function should be separated as an independent coordinate called L6, or the hypothesis of functional correspondence between human L6 and the candidate function on the LLM side.

Status:

“If the same observable distinctions can be maintained without L6, the hypothesis that L6 should be treated as an independent coordinate is weakened.”

“If other cognitive functions or combinations of them can preserve the same distinctions and L6 provides no additional distinction, the independence of L6 is weakened.”

“If LLM behavior can be organized to a similar degree without correspondence with L6, the functional correspondence hypothesis is weakened.”

“Weakening the independence of L6 is different from denying the existence of a function corresponding to the construction of explanatory models or theories.”

Q43. Why is L7 necessary as an independent coordinate?

Related Qs: Q39–Q42, Q44–Q46, Q58–Q60

L7 is treated as an independent coordinate in order to organize and infer high-abstraction, highly compressed cognition formed through accumulated experience separately from other cognitive functions.

Through repeated involvement with an object, observation, and accumulated experience, people may form sensations, insights, and highly abstract concepts that are difficult to capture simply as differences in knowledge. For example, even short expressions such as “something feels different” or “make it a little more blue” may communicate an intended meaning between people who share experience in the same field.

By treating L7 as independent, this kind of experience-based cognition can be organized separately from the construction of explanatory models or theories handled by L6.

However, L7 itself cannot be directly observed. It is hypothetically inferred from outputs appearing in Malkuth, such as speech, text, and behavior.

Status: Answerable based on the individual paper

Q44. Can L7 be replaced by other layers?

Related Qs: Q39–Q43, Q45–Q46, Q58–Q60

At present, it has not been demonstrated that L7 cannot be replaced by other layers.

In SCOPE, L7 is treated as a coordinate for organizing and inferring high-abstraction, highly compressed cognition formed through accumulated experience, and is distinguished from L6, which constructs explanatory models and theories.

However, if another layer or a combination of layers could preserve the same observable distinctions to a similar degree without L7, the need to maintain L7 as an independent coordinate would be weakened.

This would weaken the need for L7 as an independent coordinate, but would not mean that experience-based high-abstraction, highly compressed cognition itself does not exist.

Status: Answerable, but requires additional verification

Q45. What distinctions would be lost if L7 were removed?

Related Qs: Q39–Q44, Q46, Q58–Q60

If L7 is removed, the coordinate for organizing and inferring high-abstraction, highly compressed cognition formed through accumulated experience separately from other cognitive functions would be lost.

Through repeated involvement with an object, observation, and accumulated experience, people may form sensations, insights, and highly abstract concepts. These are not always constructed as explanatory models or theories, and may instead appear as short expressions or intuitive judgments.

Without L7, it would become harder to separately organize the possibility that such outputs correspond to experience-based high-abstraction, highly compressed cognition.

Therefore, what is lost by removing L7 is not the experience-based cognition itself, but the observable distinction used to organize and infer it independently.

Status: Answerable based on the individual paper

Q46. What observations would weaken the L7 hypothesis?

Related Qs: Q39–Q45, Q58–Q60

The current individual paper on L7 does not define observation-specific predictions unique to L7 or a falsification condition based on a single observation.

However, a comparative defeat condition can be set for the hypothesis that L7 should be treated as an independent coordinate.

For example, if differences in sensations, insights, and highly abstract concepts inferred to be experience-based can be distinguished and organized to the same or a greater degree using other cognitive functions or combinations of them, and adding L7 provides no additional observable distinction, the independence of L7 would be weakened.

The need for L7 as an independent coordinate would also be weakened if the same observations could be organized equally well or better by another explanation without assuming that they are experience-based.

However, what would be weakened is not the existence of experience-based high-abstraction, highly compressed cognition itself, but the hypothesis that it should be organized as an independent coordinate called L7.

The current L7 paper also does not define an operational criterion for determining what counts as “the same or a greater degree.”

Status: A comparative defeat condition can be set / L7-specific observational falsification conditions are not yet defined

Q47. Why Is L8 Necessary as an Independent Coordinate?

Related Qs: Q3, Q4, Q7–Q14, Q48–Q50, Q74–Q80, Q81–Q87, Q94–Q103

In SCOPE, L8 is treated as a coordinate for distinguishing the function of referencing the cognitive structure as a whole, including the relationships among its individual cognitive elements.

While the other layers address different cognitive functions, L8 treats the overall cognitive structure formed through their interactions as the object of analysis.

By treating L8 as independent, the function of “referencing the cognitive structure as a whole” can be maintained as a separate perspective from individual cognitive functions.

However, this does not mean that the independence of L8 has been demonstrated. Whether the same distinction can be preserved using other layers or combinations of them is examined separately in Q48.

Status: The basis for its inclusion can be explained from existing materials / its independence remains subject to comparative testing

Q48. Can L8 Be Replaced by Other Layers?

Related Qs: Q3, Q4, Q7–Q14, Q47, Q49, Q50, Q74–Q80, Q94–Q103

Current SCOPE materials do not demonstrate that L8 is irreplaceable by other layers.

SCOPE distinguishes L8 as the function that references the cognitive structure as a whole as a single structure. However, L8 is also described as functioning through interaction with other layers.

Therefore, if the same observable distinction — “referencing the cognitive structure as a whole” — can be maintained using another layer or a combination of multiple layers without positing L8, the need to retain L8 as an independent layer would be weakened.

At present, no operational criterion has been defined for making this comparison. The replaceability of L8 therefore remains an unresolved issue requiring comparative testing.

Status: Replaceability unresolved / comparative testing required

Q49. What Distinctions Would Be Lost If L8 Were Removed?

Related Qs: Q3, Q4, Q7–Q14, Q47, Q48, Q50, Q74–Q80, Q81–Q87, Q94–Q103

Removing L8 would eliminate the independent SCOPE coordinate used to represent the function of referencing the cognitive structure as a whole.

Other layers address individual cognitive functions, whereas L8 is distinguished as the function that references the cognitive structure as a whole, including the relationships among those functions. In humans, this distinction is used to describe cognition as a whole. In LLMs, it is used for the functional correspondence in which the internal structure formed through training as a whole influences output.

Removing L8 would therefore eliminate the independent coordinate used to describe and compare this function across humans and LLMs.

However, this does not mean that the distinction can only be preserved through an independent L8 coordinate. Whether another layer or combination of layers could maintain the same distinction remains the unresolved comparative question addressed in Q48.

Status: The descriptive and comparative distinction lost by removing L8 can be identified / whether that distinction is unique to L8 remains unresolved

Q50. What Observations Would Weaken the L8 Hypothesis?

Related Qs: Q3–Q5, Q7–Q14, Q47–Q49, Q74–Q80, Q94–Q103

Current SCOPE materials do not define observable predictions unique to L8 or operational falsification criteria specific to it.

At present, therefore, what can be specified is not a direct falsification condition for L8, but a comparative defeat condition for retaining it as an independent coordinate.

If a model without L8, or a combination of other layers, can preserve the same distinction of “referencing the cognitive structure as a whole,” the need to retain L8 as an independent coordinate would be weakened.

However, no operational criterion for evaluating this comparison has yet been defined. Thus, a comparative defeat condition can be stated, while an observation-based falsification criterion unique to L8 remains undefined.

Status: Comparative defeat condition can be specified / L8-specific observational falsification criterion remains undefined

Q51. Why Are Verbalization and Emotional Configuration Treated as a Slide Pair?

Related Qs: Q19–Q26, Q52–Q60

SCOPE treats L1 verbalization and L2 emotional configuration as a sliding pair.

This hypothesis is based on observations that, in states associated with strong anger or anxiety, verbalization may become more difficult or narrower in range. To organize such changes, SCOPE treats L1 and L2 not only as separate states but also as a pair with a relative relationship.

However, these observations do not by themselves demonstrate that an increase in one ratio must be matched by an equivalent decrease in the other. In the integrated SCOPE model, the sliding relationship remains a structural hypothesis.

The sliding model concerns the relative ratio within the pair for the same cognitive object, not the overall level of activation. The same ratio may occur at different overall activation levels.

Status: Structural hypothesis derived from observable output changes / placement defined in existing materials / validity of the sliding relationship itself requires further testing

Note: Evaluation of the activation concept itself is outside the scope of this question and will be addressed in later questions.

Q52. Are There Cases in Which the L1–L2 Slide Relationship Does Not Hold for the Same Cognitive Target?

Related Qs: Q19–Q26, Q51, Q53

Under the current SCOPE definition, L1 and L2 are treated as a relative ratio within a pair. Therefore, if changes involving the same cognitive object cannot be stably represented by the current sliding relationship, they would constitute candidate counterexamples to the L1–L2 hypothesis.

Examples would include cases in which a relative change corresponding to one layer is repeatedly observed without a corresponding change in the other, or cases in which both change but cannot be represented by a constant-sum sliding relationship.

However, current materials do not demonstrate that such cases are absent, nor do they provide sufficient criteria for mapping observed changes onto changes in the L1 or L2 ratio.

In addition, the sliding hypothesis cannot be preserved indefinitely by reclassifying counterexamples after the fact as changes in activation or other factors. Distinguishing ratio changes from other changes requires criteria that can be applied before a counterexample is observed.

Status: Current sliding definition established in existing materials / absence of counterexamples not demonstrated / candidate falsifying observations can be specified / criteria for mapping observations to L1 and L2 remain undefined

Q53. Would It Be More Explanatory to Treat L1 and L2 as Independent Variables?

Related Qs: Q19–Q26, Q51, Q52, Q54–Q60

That is possible. At present, SCOPE has not sufficiently compared the constant-sum sliding model of L1–L2 with a model in which L1 and L2 can vary independently.

SCOPE also distinguishes sliding ratio from overall activation. Therefore, when outputs corresponding to both L1 and L2 appear strongly, this could be interpreted either as both variables increasing independently or as the overall activation of the pair increasing while the ratio remains relatively stable.

Current materials do not provide sufficient criteria for reliably distinguishing these interpretations from observable output. For this reason, counterexamples should not be reclassified after the fact as activation changes. Criteria for distinguishing ratio change from activation change are needed in advance.

In addition, the competing model need not be complete independence. Another possibility is that L1 and L2 influence one another without maintaining a constant-sum ratio.

Therefore, the current SCOPE materials do not demonstrate that the sliding model has greater explanatory power than these competing models. Comparative testing is needed to determine whether the sliding constraint captures regularities that the alternatives do not.

Status: Independent-variable and non-constant-sum interaction models remain viable competing hypotheses / criteria for distinguishing ratio change from activation change are insufficient / superiority of the sliding model has not been demonstrated / comparative testing is required

Q54. Why Are Boundary Setting and Acceptance Treated as a Sliding Pair?

Related Qs: Q31–Q38, Q55–Q57

The L4–L5 sliding relationship was not derived deductively from theory. It was proposed as a structural hypothesis based on observations that boundary setting and acceptance can appear to change relative to one another for the same cognitive object.

SCOPE distinguishes L4 as the function that sets boundaries by judging whether an object should be let through or held apart, and L5 as the function that accepts people, information, values, theories, and other objects into cognition. Observationally, an object that was previously held apart may later be accepted, while an object that was previously accepted may later become subject to a new boundary.

However, these observations support only the possibility of a relative relationship between L4 and L5. They do not demonstrate that their ratios always sum to a constant or that an increase in one must be matched by a corresponding decrease in the other.

When both L4- and L5-related outputs appear strong, current materials also lack sufficient criteria for reliably distinguishing among higher overall activation, more independent change in L4 and L5, or differences in the cognitive object or unit being analyzed.

Status: L4–L5 sliding placement defined in existing materials / structural hypothesis derived from observed relative change / constant-sum ratio not demonstrated / criteria for distinguishing activation, independent change, and object-unit differences remain insufficient

Note: Activation is not used here either as evidence for the sliding hypothesis or as a way to avoid counterexamples.

Q55. Is It Really Impossible for Both L4 and L5 to Have High Ratios for the Same Object and Cognitive Unit?

Related Qs: Q31–Q38, Q54, Q56, Q57

Under the current SCOPE model, L4 and L5 are treated as a constant-sum ratio. Therefore, both cannot be assigned high ratios simultaneously for the same cognitive object and cognitive unit.

However, this does not demonstrate that such states do not exist in reality. Observationally, there may be cases in which a boundary is maintained while part of the same object is also being accepted.

Several explanations are possible: the overall activation of the pair may be high, the apparent single object may actually involve different cognitive objects or units, or L4 and L5 may be capable of remaining relatively high at the same time rather than obeying a constant-sum relationship.

The problem is that if such observations can always be reclassified after the fact as activation differences or object splitting, the sliding hypothesis becomes difficult to falsify. Distinguishing these possibilities requires criteria that can be applied in advance.

Therefore, if repeated observations under a fixed object and cognitive unit require both L4 and L5 to be treated as high-ratio states and cannot be stably represented by the current sliding model, the L4–L5 hypothesis would be weakened.

At present, sufficient criteria for distinguishing activation, object splitting, and genuinely high ratios in both layers have not been established. Thus, the direction of a falsification condition can be specified, but it cannot yet be applied reliably.

Status: Simultaneously high ratios are excluded by the current definition / their absence has not been demonstrated / competing explanations remain / distinguishing criteria are not established / a falsification direction can be specified but is not yet reliably applicable

Q56. To What Extent Is It Acceptable to Explain Strong L4 and L5 Responses by Saying That the Cognitive Objects Are Different?

Related Qs: Q31–Q38, Q54, Q55, Q57

In SCOPE, the L4–L5 sliding ratio is defined in relation to a specific cognitive object. Therefore, even when both L4 and L5 appear strong within the same general topic, they may actually be operating on different cognitive objects or cognitive units.

However, this explanation cannot be used freely in response to potential counterexamples. If the object can always be subdivided whenever both L4 and L5 appear strong, cases in which both are high for the same object could always be avoided.

Therefore, object splitting requires grounds independent of the sliding judgment itself. For example, even within the same event, accepting its factual content and evaluating the event may in some cases be treated as different cognitive objects.

However, current materials do not provide sufficiently stable operational criteria for determining what counts as a separate object or how finely an object may be divided.

Object splitting can therefore be considered as one competing explanation, but it cannot be adopted automatically. The object split itself must also remain subject to evaluation.

Status: Object splitting is possible / cannot be used freely to avoid counterexamples / requires independent grounds / operational criteria remain insufficient / object splitting remains one competing explanation

Q57. Does Allowing Analysts to Freely Change the Unit of Analysis Make the Sliding Model Unfalsifiable?

Related Qs: Q31–Q38, Q54–Q56, Q58–Q60, Q94–Q103

Yes, that risk exists. The problem is not object splitting itself, but changing the unit of analysis after observing the result and then treating only the revised analysis as a successful continuation of the same test.

If an object contains multiple cognitive units, separating them for analysis may be appropriate. However, changing the unit requires grounds independent of whether the sliding hypothesis succeeds or fails. “The sliding relationship only works after splitting the object” is not sufficient justification by itself.

If the unit is changed, the test under the original unit ends there, and the revised analysis must be treated as a separate test. The result obtained under the original unit must not be erased.

If the conclusion changes depending on how the unit is divided, and there is no independent basis for choosing among those units, the observation becomes indeterminate. In particular, if the sliding relationship holds only after post hoc redivision, falsifiability has not been secured.

Therefore, a procedure that allows the unit of analysis to be freely changed cannot by itself constitute an observational test of the sliding hypothesis. At present, prior operational criteria for unit selection have not been established.

Status: Unit changes are not prohibited / revised analyses must be treated as separate tests / original results must be retained / indeterminate when unit choice lacks independent grounds / post hoc redivision does not secure falsifiability / prior criteria remain undefined

Q58. Why are theory construction and expQ58. Why Are Theorization and Experience-Derived High-Abstraction Cognition Treated as a Sliding Relationship?

Related Qs: Q39–Q46, Q59, Q60

SCOPE distinguishes L6 as the function that constructs explanatory models and theories, and L7 as high-abstraction, highly compressed cognition formed through accumulated experience. For the same cognitive object, the relative tendency to rely more strongly on one or the other is hypothesized as the L6–L7 sliding ratio.

This relationship was not introduced from theoretical symmetry. It was derived from observations of relative shifts between developing an object into an explanatory model and relying on cognition compressed from accumulated experience.

However, deriving the hypothesis from observation does not itself establish the validity of the sliding relationship. The fact that L6 and L7 can be functionally distinguished also does not demonstrate that they must be represented as a single ratio.

In the current SCOPE model, L6 and L7 are treated not as two independent variables but as a ratio within one pair. This ratio does not represent activation itself. Therefore, merely observing both functions strongly does not constitute a counterexample to the sliding hypothesis.

If repeated observations of the same object and the same cognitive unit cannot be represented as a sliding relationship, the hypothesis would be weakened. Whether treating L6 and L7 as independent variables provides a better account also requires comparison. However, no prior criterion has yet been established for distinguishing ratio from activation on the basis of observable output, so such comparison cannot currently be conducted adequately.

Therefore, the L6–L7 slide is retained as a relational hypothesis derived from observation, while the necessity of representing the relationship specifically as a slide remains unestablished.

Status: functional distinction established / slide derived from observed relative changes / not theory-first / observational origin ≠ evidence of validity / activation and ratio are distinct / both appearing strong is not by itself a counterexample / discrimination criteria remain unestablished / necessity of the slide remains unestablished

Q59. How Should a State Be Treated as a Ratio When Both Theoretical Construction and Experience-Based Cognition Are Strongly Referenced?

Related Qs: Q39–Q46, Q57, Q58, Q60

In SCOPE, the L6–L7 slide does not represent activation itself. It represents the ratio within the pair: which of L6 or L7 is referenced more strongly relative to the other for the same cognitive object.

Therefore, even if both theoretical construction and experience-derived cognition appear strongly in the output, this does not mean that both are at a high ratio. In principle, both may

have high activation while the within-pair ratio remains L6-leaning, L7-leaning, or near the center.

However, a state in which both appear strong cannot automatically be explained as a matter of activation. At present, no prior criterion has been established for distinguishing activation from sliding ratio on the basis of output. It is also possible that different objects or cognitive units are involved, but units cannot be divided automatically in order to avoid a counterexample. The conditions for dividing cognitive objects are addressed in Q57.

Therefore, when both appear strong for the same object and the same cognitive unit, at least the following competing hypotheses should be retained:

- Both L6 and L7 have high activation while the sliding ratio is preserved
- L6 and L7 are better treated as independent variables rather than as a sliding ratio
- Different objects or cognitive units are actually included

If these cannot be distinguished from observable output, it should not be concluded either that the slide holds or that it has failed. The observation should be treated as indeterminate.

If future conditions allow activation and ratio to be distinguished for the same object and the same cognitive unit, and independent changes that cannot be represented by a sliding ratio are consistently observed, the L6–L7 slide hypothesis would be weakened. At present, the necessary discrimination criteria have not been established.

Therefore, a state in which both theoretical construction and experience-derived cognition are strongly referenced does not immediately contradict the current SCOPE model. However, it has also not been verified that such a state can be explained by activation. Until discrimination criteria are established, multiple interpretations must be retained.

Status: both appearing strong is not by itself a counterexample / both are not therefore classified as high ratio / automatic explanation by activation is prohibited / criteria for distinguishing ratio from activation are not established / separation into different objects or cognitive units remains a competing hypothesis / automatic splitting is prohibited / indeterminate when hypotheses cannot be distinguished / weakening condition applies after discrimination criteria are established / comparison with an independent-variable model is addressed in Q60

Q60. How Can the Sliding Model Be Shown to Be Superior to Treating L6 and L7 as Independent Coordinates?

Related Qs: Q39–Q46, Q53, Q57–Q59

At present, it has not been established that treating L6 and L7 as a sliding relationship is superior to treating them as two independent variables.

SCOPE hypothesizes L6 and L7 as a ratio within a pair because relative changes in the cognition corresponding to the two have been observed. However, deriving a hypothesis

from observation is separate from demonstrating that it has greater explanatory power than competing models.

A valid comparison would require applying the sliding model and competing models to the same cognitive object, cognitive unit, and set of outputs under the same conditions. If the object or unit is changed after observing the results, this should be treated as a separate comparison rather than a continuation of the original one.

Competing models are not limited to models in which L6 and L7 are completely independent variables. A model in which they influence one another without forming a fixed sliding relationship is also a relevant competitor.

One form of support for the sliding model would be that, under identical conditions, it consistently explains the observations as a single relative relationship with fewer exception rules or additional assumptions than competing models. However, fewer assumptions alone do not establish superiority, and greater model flexibility does not itself mean greater explanatory power. Conversely, if a competing model consistently explains changes that are difficult to represent as a slide, the necessity of the sliding model would be weakened.

At present, however, no prior criterion has been established for distinguishing ratio from activation on the basis of observable output. It therefore remains unclear whether a state in which both appear strong reflects high activation while preserving the ratio, or a more independent change in L6 and L7. Without this distinction, results could be assigned to either explanation after observation, preventing a fair comparison.

Therefore, the current claim is limited to the L6–L7 slide being a relational hypothesis derived from observation. Demonstrating superiority over competing models would require criteria for distinguishing ratio from activation, followed by comparative testing using the same conditions and the same data.

Status: superiority over competing models is unestablished / observational origin ≠ comparative superiority / comparison requires the same object, unit, and data / changing the unit afterward constitutes a separate comparison / competitors are not limited to fully independent-variable models / fewer additional assumptions alone do not establish superiority / flexibility ≠ explanatory power / criteria for distinguishing ratio from activation are unestablished / conditions for fair comparison are not yet available / necessity of the sliding model is unestablished

Q61. Why Do These Connection Lines Exist Between the Current Layers?

Related Qs: Q9, Q12–Q14, Q58–Q60, Q62–Q66

Here, “connection lines” refers to the lines shown in the current SCOPE diagram.

In SCOPE, a connection line does not represent processing order or a fixed pathway of information flow. It represents a relationship in which two layers may influence one another.

Not all connections are assumed to exist for the same reason; the basis for each connection may differ by pair.

SCOPE as a whole was not created by first designing a symmetrical diagram and then fitting human cognition into it. Rather, recurring relationships observed in cognitive output were provisionally placed as hypotheses and organized into a cognitive coordinate system.

However, three points must be distinguished:

- the definition that a connection line represents a possible mutual influence,
- the historical fact that the connection structure was hypothesized from observed relationships,
- and the validity of claiming that each specific pair requires the connection shown.

The fact that a relationship was derived from observation does not establish the validity of every connection in the current diagram. Likewise, the possibility that two layers influence one another does not by itself demonstrate that they require a direct connection.

Therefore, what is currently established is only that the present connection diagram represents a hypothesized organization of observed cognitive relationships. The necessity of each individual connection, and the validity of the connection structure as a whole, remain unestablished and must be evaluated separately from their definition.

Status: connection lines do not represent processing order or fixed pathways / they represent possible mutual influence / reasons for connections may differ by pair / connection structure was hypothesized from observation / observational origin $\neq$ validity of each connection / possible interaction $\neq$ necessity of direct connection / necessity of individual lines remains unestablished / validity of the overall connection structure remains unestablished / definition and validity must be distinguished

Q62. How Are Direct and Indirect Connections Distinguished?

Related Qs: Q9, Q12–Q14, Q61, Q63–Q66

At present, SCOPE has no established operational criterion for distinguishing direct from indirect connections on the basis of observable output.

Connection lines in SCOPE do not represent processing order or fixed causal pathways. They represent relationships in which layers may influence one another. Therefore, the fact that two layers are connected by a line in the diagram is not equivalent to showing that one directly influences the other.

Because the internal layers themselves cannot be directly observed, simultaneous or sequential changes corresponding to two layers do not by themselves demonstrate a direct influence between them. A third layer may be involved in both, or multiple relationships may be operating at the same time.

Thus, the current diagram can distinguish graphically between adjacent pairs connected by a line and non-adjacent pairs connected through multiple lines. However, this graphical distinction cannot be equated with direct and indirect influence within cognition.

To distinguish direct from indirect influence theoretically, separate observational criteria capable of discriminating between them would be required. Such criteria have not yet been established.

Status: connection lines do not represent direct causal pathways / graphical adjacency is distinct from direct internal influence / internal layers are not directly observable / simultaneous or sequential change $\neq$ evidence of direct connection / third-layer and other competing hypotheses remain / operational criteria for distinguishing direct and indirect influence are unestablished

Q63. If Removing a Single Connection Line from the Current Diagram Does Not Change the Analysis, Is That Connection Necessary?

Related Qs: Q3, Q5, Q13–Q14, Q61–Q62, Q64–Q66

Here, “removing one line” means removing only the direct connection between a particular pair of layers in the current diagram and testing whether the same analysis still holds under that altered structure. This comparison should be performed one connection at a time across the current diagram.

If another pathway that already existed in the diagram before the comparison remains, the two layers do not become completely unrelated in the diagram. If the removed line was the only connection of a layer, however, that layer may become isolated.

To evaluate the analytical necessity of a connection, the structure containing the line and the structure with only that line removed must be compared using the same observational material, inference procedure, and judgment criteria. Any alternative pathway used in the comparison must already exist in the current diagram before the comparison begins.

If removing the line makes previously distinguishable competing hypotheses indistinguishable, or produces a reproducible difference in inference results or discriminability, this may support the claim that the direct connection has an analytical role.

Conversely, if removing the line produces no difference, the necessity of that direct connection is not supported under those conditions. However, this alone does not show that the line is generally unnecessary. The connection may be redundant, or its role may simply be undetectable with the current observations or analytical methods.

The fact that removing the line makes the direct connection unavailable in the diagram is only a definitional consequence and does not itself demonstrate necessity. Likewise, observing two layers as active at the same time does not establish that a direct connection between them is required.

New pathways or explanations must not be added after the comparison in order to reproduce the original result. The comparison conditions must be fixed before the outcome is observed.

At present, the analytical necessity of each connection in the current SCOPE diagram has not been systematically tested through this type of one-line deletion comparison. Therefore, the fact that a connection is present in the diagram must be distinguished from evidence that the direct connection is analytically necessary.

Status: presence of a connection $\neq$ analytical necessity / one-line deletion = removal of only the direct connection between two layers / pre-existing alternative pathways may be used / removing a unique connection may isolate a layer / compare line-present and line-removed structures under identical conditions / reproducible differences after deletion may support an analytical role / no difference leaves redundancy or non-discriminability as possibilities / simultaneous involvement $\neq$ necessity of direct connection / post hoc addition of explanations or pathways is prohibited / necessity of individual connections has not been systematically tested

Q64. Could a Different Connection Diagram Explain the Same Phenomena to a Similar Degree?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q63, Q65–Q66

Yes, that is possible.

Q63 addressed the case of removing a single connection from the current diagram. Q64 instead asks whether an alternative diagram, while retaining the same layers, could explain the same observational material equally well by adding, removing, or changing only the connection pattern.

The fact that the current diagram represents hypothesized cognitive relationships does not mean that only this configuration is viable. If an alternative diagram produces the same inference results and preserves the same ability to distinguish competing hypotheses, the current configuration cannot be uniquely supported by those observations alone.

More connections also do not necessarily improve analytical power. Adding new direct connections may make hypotheses that were previously distinguishable harder to separate.

Comparisons with alternative diagrams must use the same observational material, inference procedure, and judgment criteria, and the diagrams to be compared and the meaning of “equally well” must be defined before the results are examined. The comparison should consider not only inference results but also how well competing hypotheses can be distinguished.

If an alternative diagram produces the same inference results and comparable discriminability, this does not immediately show that the current diagram is wrong. However,

it weakens the claim that the current configuration is uniquely necessary, leaving multiple diagrams as competing hypotheses that cannot yet be distinguished.

Conversely, if hypotheses distinguishable under the current diagram become indistinguishable under an alternative diagram, or if reproducible differences emerge between them, those differences may provide evidence for comparing connection structures. This still does not mean that the actual internal cognitive pathways have been directly observed.

New explanations, auxiliary hypotheses, or connections must not be added after the comparison in order to reproduce the original results. The comparison conditions must be fixed in advance.

At present, the current SCOPE connection diagram has not been systematically compared with multiple alternative configurations under identical conditions in a way that establishes its superiority. Therefore, the current diagram should be treated as the presently proposed structural hypothesis, while the possibility that alternative connection structures could explain the same phenomena equally well remains an unresolved competing hypothesis.

Status: presence of current diagram $\neq$ uniqueness of its configuration / compare alternative connection patterns while retaining the same layers / additions, deletions, and changes of connections are included / more connections $\neq$ greater analytical power / altered wiring may reduce discriminability / criteria for “equally well” must be defined in advance / equivalent alternatives weaken the claim of unique necessity / alternative viability $\neq$ immediate rejection of current diagram / reproducible differences may inform comparison / does not directly reveal actual internal pathways / post hoc explanations, auxiliary hypotheses, and added connections are prohibited / superiority of the current configuration remains unestablished

Q65. If Layers Can Simply “Influence One Another,” Why Not Fully Connect All Layers?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q64, Q66–Q67

That possibility cannot currently be excluded.

Here, a fully connected model means retaining the same set of layers, L0 through L8, while adding a direct connection between every pair of layers in the diagram.

If a connection line means only that two layers may influence one another, the rationale for leaving some pairs without direct connections becomes weaker. A fully connected diagram must therefore be considered a competing hypothesis to the current configuration.

If the fully connected model does not require every added connection to be used, it may be able to reproduce the same analytical results as the current diagram. If the added connections do not affect inference results or the ability to distinguish competing hypotheses, the analytical necessity of the current non-connections would be weakened.

However, more connections do not necessarily mean greater analytical power. Allowing direct connections between every pair removes the distinction between pairs that are directly connected and pairs that are not, which may reduce the ability to distinguish competing hypotheses.

Therefore, the current diagram and the fully connected model should be compared using the same observational material, inference procedure, and judgment criteria, evaluating not only inference results but also discriminability between competing hypotheses.

If the fully connected model produces the same inference results and comparable discriminability, the necessity of the current non-connections would be weakened. Conversely, if full connectivity reduces discriminability or produces reproducible differences, this may support the analytical role of a non-fully-connected structure.

Even then, however, this would not establish that the current specific connection pattern is uniquely valid. The usefulness of a non-fully-connected structure and the uniqueness of the current configuration are separate questions.

At present, the current SCOPE diagram has not been systematically compared with a fully connected model. The fully connected model therefore remains an unresolved competing hypothesis for testing the necessity of the current connection structure.

Status: possible mutual influence $\neq$ necessity of direct connection / fully connected = direct connections between all L0–L8 pairs / possible mutual influence alone does not exclude a fully connected model / full connectivity may reproduce current results / more connections $\neq$ greater analytical power / full connectivity may reduce discriminability / compare current and fully connected models under identical conditions / equivalent performance weakens the necessity of non-connections / reduced discriminability under full connectivity may support a non-fully-connected structure / usefulness of non-full connectivity $\neq$ uniqueness of current configuration / superiority of the current diagram remains untested

Q66. What Is the Theoretical Meaning of Pairs That Have No Connection Line?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q65, Q67

In the current SCOPE model, the absence of a connection line means that the two layers are not represented as directly connected in the diagram.

This does not mean that the two layers are unrelated, cannot influence one another, or cannot be involved at the same time. The only direct implication of a non-connection is that the current diagram does not represent the pair as a direct relationship.

This distinction may allow different structural hypotheses to be retained separately. For a given observation, one hypothesis may treat two layers as directly connected, while another may involve relationships with other layers. For example, because L3 is connected to all

layers, two layers that are not directly connected in the current diagram can still be analyzed through hypotheses involving their respective relationships with L3.

Thus, non-connections serve a configurational purpose: they allow different structural hypotheses to remain distinct rather than collapsing them into a single direct relationship. However, not all non-connections are necessarily based on the same rationale, and each must be examined individually.

The existence of this purpose does not itself demonstrate analytical necessity. If adding a direct connection produces no difference in inference results or in the ability to distinguish competing hypotheses, the necessity of that non-connection would be weakened. Conversely, if adding the line reduces discriminability or produces reproducible differences, this may support an analytical role for the non-connection.

A non-connection also must not be interpreted as claiming that cognition necessarily passes through another layer internally. Graphical non-connection and actual internal causal pathways must be kept distinct.

Therefore, non-connections in the current SCOPE model should be treated as configurational distinctions that help preserve alternative structural hypotheses. However, the necessity of each specific non-connection, and the superiority of the current non-connection pattern over alternatives, have not yet been systematically established.

Status: non-connection $\neq$ no relationship / non-connection $\neq$ no mutual influence / non-connection $\neq$ inability to co-occur / means the pair is not represented as directly connected / preserves distinct structural hypotheses / hypotheses involving L3 may remain separate / rationales for non-connections need not all be identical / intended distinction $\neq$ demonstrated analytical necessity / no difference after adding a line weakens necessity / reduced discriminability or reproducible differences may support a role / non-connection $\neq$ necessary internal mediation through another layer / necessity of specific non-connections and superiority of the current pattern remain unestablished

Q67. Why Is L3 the Only Layer Connected to All Other Layers?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q66, Q68–Q73

In the current SCOPE model, L3 is connected to all other layers because it is positioned as the layer responsible for internal self-consistency.

Here, self-consistency does not mean correspondence with reality or truth evaluation. In humans, it refers to organizing values, beliefs, worldviews, self-image, and related elements into a form with relatively little internal contradiction. In LLMs, SCOPE does not assume an identical internal mechanism, but treats maintenance of response-level coherence and basic system-level policies as possible functional correspondences.

L3's full connectivity does not mean that L3 is a central command unit, that all cognition must pass through L3, or that L3 controls other layers in one direction. L3 itself functions through interaction with other layers and may also be updated.

In the current SCOPE model, the function corresponding to self-consistency is not restricted to relations with only a subset of cognitive functions. L3 is therefore positioned so that its relation to every layer can be represented. Cognition formed in different layers does not necessarily point in the same direction, and the relation between those forms of cognition and existing self-consistency may affect how the overall cognitive structure is organized.

Thus, L3's full connectivity represents the structural hypothesis that self-consistency may relate to every layer.

However, it has not been demonstrated that a structure in which only L3 is fully connected is uniquely valid. Alternative configurations in which another layer is fully connected, L3 is not central, or a different connection structure is used may also explain the same observations.

Therefore, the current claim is limited to this: SCOPE adopts full connectivity for L3 in order to represent self-consistency as a function that may interact with all layers. The uniqueness of this configuration and its superiority over alternative structures remain untested.

Status: full connectivity of L3 $\neq$ central command unit / $\neq$ all cognition must pass through L3 / $\neq$ one-way control / L3 does not determine truth or correspondence with reality / human and LLM cases are treated as functional correspondence rather than identical mechanisms / self-consistency functions through interaction with other layers / L3 itself may be updated / full connectivity represents the structural hypothesis that self-consistency may relate to every layer / necessity of making only L3 fully connected is unproven / superiority over alternative structures remains untested / non-L3-centered alternatives are examined in Q69

Q68. Can Cognitive Direction Be Formed Without L3?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q67, Q69–Q73

Yes.

However, “without L3” does not mean that L3 completely stops functioning or that an internal pathway bypassing L3 has been directly observed. Because SCOPE cannot directly observe internal layers, this question includes cases in which functions corresponding to other layers strongly shape output or behavior while the influence corresponding to L3 is not sufficiently reflected.

In the current SCOPE model, L3 is neither a fixed starting point nor a mandatory passage point for cognition. Cognitive structure is formed through interaction among the layers.

For example, in situations where survival or life maintenance is strongly required, functions corresponding to L0, such as threat avoidance and survival prioritization, may become dominant. In such cases, values, beliefs, self-image, and other elements normally inferred as corresponding to L3 may not be sufficiently reflected in immediate output or behavior.

This does not mean that L3 has disappeared. Nor is stronger or weaker reflection of L3 itself treated as a judgment of morality or correctness.

After the crisis has passed, self-consistency corresponding to L3 may again become more strongly reflected, and a mismatch with earlier behavior may be recognized. However, such post-event reactions cannot be attributed to L3 alone.

Therefore, SCOPE does not assume that L3 must always precede the formation of cognitive direction. At the same time, this does not demonstrate that a structure in which L3 is not a necessary condition can by itself explain the formation of the entire cognitive structure. That question requires separate examination.

Status: L3 is not a fixed starting point or mandatory passage point / “without L3” ≠ complete shutdown of L3 / ≠ direct observation of an internal bypass / L3 may be insufficiently reflected in output / L0-related functions may dominate in survival situations / degree of L3 reflection ≠ morality or correctness / cognition and behavior may form without strong reflection of L3 / self-consistency may become more strongly reflected after a crisis / post-event reactions cannot be attributed to L3 alone / whether the whole cognitive structure can be explained without treating L3 as necessary remains unresolved

Q69. Could the same phenomena be explained by an alternative model that does not centralize self-consistency in L3?

Related Qs: Q3–Q5, Q13–Q14, Q61–Q68, Q70–Q73

Possibly.

In the current SCOPE model, the function corresponding to self-consistency is assigned to L3 and represented as a fully connected structure capable of relating to all other layers. However, this does not mean that this arrangement is the only one capable of explaining the observed phenomena.

For example, the same observations might also be explained by a distributed model in which self-consistency emerges from interactions among multiple cognitive functions, or by a model in which local forms of coherence arise between cognitive functions and collectively produce overall consistency.

What SCOPE can directly observe is output appearing in Malkuth, such as speech, text, and behavior, rather than the internal structure itself. Therefore, if the current L3 arrangement and an alternative model explain the same observations equally well, those observations alone cannot distinguish between them.

Accordingly, supporting the placement of L3 requires more than showing that the phenomena can be explained by introducing L3. It is necessary to identify differences between competing models, such as whether they generate different predictions, differ in the range of observations they can explain, or require different amounts of additional assumptions.

At present, systematic comparison with alternative models that do not centralize self-consistency in L3 has not been sufficiently conducted. The fully connected L3 structure should therefore be regarded as a structural hypothesis adopted within the current SCOPE model, rather than as an arrangement whose uniqueness or superiority over alternatives has been empirically established.

However, the mere existence of an alternative model capable of explaining the same observations does not immediately falsify the current SCOPE model. If the models are observationally indistinguishable, what can be concluded is only that the current arrangement cannot be regarded as the uniquely supported explanation. Future work must identify observations and predictive differences that could distinguish between them.

Status: Alternative models that do not centralize self-consistency in L3 have not been excluded / explanatory success from adding L3 $\neq$ proof that the L3 arrangement is correct / distributed or locally coherent structures remain possible competing hypotheses / observationally indistinguishable when the same observations are explained equally well / the current L3 arrangement remains a structural hypothesis / its uniqueness and superiority over alternatives have not been established / the existence of an alternative model alone does not immediately falsify SCOPE / comparison through predictive differences, explanatory range, and additional assumptions is required

Q70. How does SCOPE handle cognition when self-consistency is weak, unformed, or contradictory?

Related Qs: Q27, Q29, Q30, Q67–Q73, Q94, Q103

SCOPE does not treat the inability to infer a single, stable form of L3 self-consistency as equivalent to the failure of cognition itself.

L3 addresses the organization of values, beliefs, worldviews, self-image, and related elements into a relatively consistent form within the individual. However, this self-consistency does not guarantee correspondence with reality or logical correctness, and multiple forms of self-consistency may coexist or change over time.

Therefore, contradictory output alone is not sufficient to conclude that “L3 is weak” or “L3 is underdeveloped.” Multiple forms of self-consistency may exist for different objects or situations, or the observed contradiction may result from the influence of other layers on output.

In addition, “weak” or “underdeveloped” self-consistency is not currently defined as an independent state within SCOPE, and no criteria for identifying such a state have been established. If such a concept is used, it should be treated as a hypothesis requiring additional validation rather than as part of the existing definition.

At present, it has not been shown that observation alone can uniquely distinguish between multiple coexisting forms of self-consistency, insufficiently formed self-consistency, and contradictions produced by the influence of other layers. When these possibilities cannot be

distinguished, SCOPE retains them as competing structural hypotheses rather than forcing a single interpretation.

Status:

Contradictory output or the inability to infer a single stable form of self-consistency does not imply failure of L3 or cognition / multiple forms of self-consistency may coexist within L3 / “weak” or “underdeveloped” self-consistency is not a currently defined SCOPE state / when observations cannot distinguish among competing explanations, they are retained as competing structural hypotheses

Q71. Can the “update cost” of L3 be estimated independently?

Related Qs: Q27, Q30, Q67–Q73, Q94, Q103

At present, the “update cost” of L3 cannot be estimated as an independent single variable.

Here, update cost is a hypothetical concept describing how much readjustment may be required across the broader cognitive structure and its relationships with other layers when the self-consistency maintained by L3 is changed. It does not simply mean the “strength” of L3.

When the required range of readjustment is large, a hypothesis may be placed that the relevant self-consistency is relatively difficult to change.

The fact that an analyst revises a hypothesis about L3 does not itself mean that the target’s self-consistency has changed.

The individual L3 paper states that self-consistency is not always fixed and may change when contradictions with observed facts or experiences become sufficiently large. At the same time, such change does not necessarily correspond directly to the magnitude of an event. A relatively minor event may lead to change, while an existing self-consistency may remain even when substantial counterevidence is presented. The process through which such change occurs also requires consideration of relationships with other layers and cannot be sufficiently explained by L3 alone.

Accordingly, when strong rejection or resistance concerning a particular target is repeatedly observed, this may serve as a clue for placing a hypothesis that the range of readjustment required for change is large. However, strong rejection or resistance itself cannot be treated as identical to update cost. Such responses may also involve processes corresponding to other layers, the cognitive object, or contextual factors.

Conversely, even if an output changes quickly in response to new information or counterevidence, this alone does not establish that update cost is low. Acquiring new knowledge or experience is not the same as changing self-consistency itself.

When placing a hypothesis about update cost, the analysis may consider the hypothesis concerning the relevant self-consistency, observed responses to information or experiences that are inconsistent with it, and its relationships with other relevant cognitive hypotheses. If

change appears to require readjustment across a broad range, a hypothesis of relatively high update cost may be placed. However, this does not constitute an independent measurement procedure or operational indicator.

Using multiple observations does not, by itself, require the description to be treated as COS. A description that uses multiple observations as evidence to compare and infer hypotheses about self-consistency may still be treated as SCOPE.

By contrast, a process that tracks the maintenance or change of self-consistency as the continual updating of a model of the target is treated as an interface between SCOPE and COS rather than as SCOPE alone. However, the accumulation of multiple observations or the length of time involved is not, by itself, a sufficient condition for treating the process as COS.

At present, no prior criterion has been specified that can uniquely distinguish, from observation alone, between revision of the analyst's inference, change in the target's self-consistency, and processing updates within COS. When these cannot be distinguished, the interpretation should not be fixed to any one of them.

The terms "high" and "low" do not refer to an independent numerical scale. Even when a hypothesis of high update cost is placed for a particular self-consistency, it cannot be generalized as a fixed characteristic of the cognitive subject as a whole. The self-consistencies handled by L3 may range from everyday preferences to values, beliefs, worldviews, and self-image, and their susceptibility to change may differ depending on the target.

Therefore, at present, update cost is not a variable that can be directly and independently estimated from L3. It should instead be treated as an auxiliary concept for placing a hypothesis that the range of readjustment accompanying change may be relatively large or small, using the relevant self-consistency, the surrounding cognitive structure, and observed responses to demands for change or contradictions as clues.

The current SCOPE framework does not define a scale for measuring update cost as an independent numerical value, nor does it define an operational indicator for changing another person's self-consistency.

Status: Update cost is a hypothetical auxiliary concept describing the range of readjustment accompanying change / Cannot be estimated as an independent single variable from L3 / Relative hypotheses of higher or lower update cost may be placed from observed responses, but cannot be confirmed / Revising the analyst's L3 hypothesis does not imply a change in the target's self-consistency / Accumulation of multiple observations or passage of time alone does not make a description COS / Tracking a process as continual updating of a target model lies at the SCOPE–COS interface / No prior criterion for uniquely distinguishing revision of the analyst's inference, change in the target's self-consistency, and processing updates within COS has been specified / Target-dependent rather than a fixed characteristic of the cognitive subject as a whole / No independent numerical or operational measurement criterion has been defined

Q72. What Criteria Can Be Used to Judge That a Counterexample Changed Other Layers Without Changing L3?

Related Qs: Q27, Q30, Q67–Q73, Q94, Q103

Current SCOPE does not define an independent criterion that can directly determine that “L3 was maintained while only other layers changed.”

Even when an input is inconsistent with the existing L3, other layers may adjust while L3 is maintained—for example, L4 may separate or exclude it, L5 may accept it only to a limited extent, or L6 may construct an explanatory structure that remains compatible with the existing L3. Conversely, if placing a different self-consistency hypothesis produces greater consistency with the cognitive structure as a whole, the possibility of an L3 update may also be considered.

Accordingly, at the observational level, both “L3 maintenance + adjustment of other layers” and “L3 update” are treated as structural hypotheses, and their consistency with the observed output and the cognitive structure as a whole is compared. If outputs continue to appear that are consistent with a previously placed self-consistency hypothesis, an L3-maintenance hypothesis may be considered. If the observations are difficult to reconcile with the previous self-consistency hypothesis, an L3-update hypothesis may be considered.

However, the fact that the analyst revises the inferred hypothesis about L3 does not itself mean that the target’s self-consistency has changed.

This comparison is also not a definitive decision rule. An output may change while the change can still be explained only by changes in other layers. Conversely, L3 may have changed even when similar outputs continue to appear.

If both hypotheses remain consistent with the observations and there is insufficient basis for preferring one over the other, neither should be fixed as the sole interpretation. They should instead be retained as competing hypotheses.

Using multiple observations does not, by itself, require the description to be treated as COS. A description that uses multiple observations as evidence to compare and infer an L3-maintenance hypothesis and an L3-update hypothesis may still be treated as SCOPE.

By contrast, a process that tracks the maintenance or change of self-consistency as the continual updating of a model of the target is treated as an interface between SCOPE and COS rather than as SCOPE alone. However, the accumulation of multiple observations or the length of time involved is not, by itself, a sufficient condition for treating the process as COS.

At present, no prior criterion has been specified that can uniquely distinguish, from observation alone, between revision of the analyst’s inference, change in the target’s self-consistency, and processing updates within COS. When these cannot be distinguished, the interpretation should not be fixed to any one of them.

Therefore, current SCOPE does not provide an independent operational criterion that uniquely distinguishes “L3 maintenance + adjustment of other layers” from “L3 update.” Multiple structural hypotheses can be compared using observed outputs and their consistency with the cognitive structure as a whole, but that comparison itself cannot be treated as a definitive decision rule.

Status: “L3 maintenance + adjustment of other layers” and “L3 update” are both possible structural hypotheses / No independent operational criterion for uniquely distinguishing them has been defined / Consistency with observed output and the cognitive structure as a whole is a comparative clue, not a definitive rule / Revising the analyst’s L3 hypothesis does not imply a change in the target’s self-consistency / When there is insufficient basis for preferring one hypothesis, competing hypotheses are retained / Accumulation of multiple observations or passage of time alone does not make a description COS / Tracking a process as continual updating of a target model lies at the SCOPE–COS interface / No prior criterion for uniquely distinguishing revision of the analyst’s inference, change in the target’s self-consistency, and processing updates within COS has been specified

Q73. What observations would weaken the hypothesis that L3 plays a central role in shaping cognitive direction?

Related Qs: Q27–Q30, Q67–Q72, Q94, Q103

In SCOPE, L3 is not positioned as the starting point of cognition or as a central control mechanism. Rather, it is treated as a layer that interacts with other layers, contributes to consistency with self-consistency, and may influence the direction in which the cognitive structure converges.

This hypothesis would be weakened if placing a self-consistency hypothesis in L3 repeatedly failed to produce any meaningful difference in organizing or distinguishing observed cognitive directions.

For example, if cognitive direction repeatedly changes through changes corresponding to other layers while the L3 hypothesis remains fixed, and changing the L3 hypothesis still produces no clear difference in how the observations are organized, the hypothesis of L3’s central involvement would be weakened.

The hypothesis would also be weakened if an alternative model that does not place L3 at the center can organize the same observations equally well or better.

A single output that does not fit the hypothesized L3 self-consistency, however, is not sufficient. Because L3 functions through interaction with other layers, an isolated mismatch in output does not by itself rule out a central role for L3.

What matters, therefore, is whether the organizational or discriminative advantage of placing L3 at the center repeatedly fails to appear across multiple observations and changes in the hypothesized L3 structure.

Current SCOPE does not define an independent operational measure of “central involvement” or a numerical threshold for rejecting this hypothesis. Determining how much evidence is required for rejection remains a matter for future validation.

Status:

The central-involvement hypothesis is weakened if placing L3 at the center repeatedly provides no organizational or discriminative advantage / changing the L3 hypothesis without producing a meaningful difference also weakens it / an alternative model that performs equally well or better also weakens the hypothesis / a single mismatch is not sufficient for falsification / no operational criterion or rejection threshold for central involvement is currently defined in SCOPE

Q74. Is it really necessary to distinguish L8 from Da'at?

Related Questions: Q75–Q80

In the current SCOPE model, L8 and Da'at are defined as separate concepts.

L8 is a cognitive layer that references the cognitive structure as a whole and judges whether that whole can, at that point, form a single judgment or understanding. However, the current SCOPE model does not define independent criteria for what constitutes “capable of forming” such a judgment or understanding.

Da'at, by contrast, is not a layer. It is an integration point representing the state in which the cognitive structure has come together as a single judgment or understanding.

This distinction allows SCOPE to describe separately “the function of judging whether the cognitive structure as a whole can form a single judgment or understanding” and “the state in which it has come together as a single judgment or understanding.” It therefore allows a distinction between a condition in which integration is progressing but has not yet come together as a single judgment or understanding, and a condition in which it has.

However, the descriptive usefulness of this distinction does not mean that L8 and Da'at must necessarily be maintained as separate concepts.

At present, there is therefore a descriptive basis for distinguishing L8 from Da'at, but it has not been established that this separation is theoretically indispensable or that this is the only possible model.

Status:

The descriptive distinction can be made under the existing definitions / criteria for L8's judgment of possible formation remain undefined / necessity and uniqueness remain unconfirmed / alternative models remain possible

Q75. What Is the Difference Between L8 Judging Something as “Capable of Forming” and Integration Being Established as Da'at?

Related Questions: Q74, Q76–Q80

In the current SCOPE model, L8 and Da'at are treated as distinct concepts.

L8 is a cognitive layer that references the cognitive structure as a whole and judges whether that whole can, at that point, form a single judgment or understanding.

Da'at, by contrast, is not a layer that performs cognitive processing. It is an integration point representing the state in which the cognitive structure has actually come together as a single judgment or understanding.

Therefore, L8's judgment that something is “capable of forming” concerns whether the cognitive structure can be brought together as a single judgment or understanding, whereas the establishment of Da'at concerns the state in which it has actually come together as such.

In humans, integration may be progressing even though cognition has not yet come together as a single judgment or understanding, and this intermediate condition may sometimes appear in Malkuth through speech or bodily reactions. In the current SCOPE model, such output may be described as occurring before Da'at has emerged.

Moreover, the emergence of Da'at does not itself constitute external output. In the current SCOPE model, what has come together at Da'at is organized as appearing in Malkuth only after passing through processing related to transmission.

In LLMs, by contrast, the moment at which a single generation becomes established is treated as functionally corresponding to the emergence of Da'at.

However, the current SCOPE model does not define what criteria L8 uses to judge something as “capable of forming,” nor does it define independent conditions for identifying the boundary between that judgment and the establishment of Da'at.

Status:

Conceptually distinct / criteria for L8's judgment of possible formation are undefined / independent criteria for identifying the boundary with the establishment of Da'at are also undefined

Q76. Is Da'at simply a name given to the state in which a judgment or understanding has been formed?

Related Qs: Q74, Q75, Q77–Q80

Under the current definition of SCOPE, this criticism is partly correct.

Da'at is not defined as an independent cognitive layer or function that performs cognitive processing. It is a concept representing the state in which multiple cognitive structures have been integrated into a single judgment or understanding. In that sense, Da'at can indeed be described as a name given to the state in which a judgment or understanding has been formed.

However, this term allows SCOPE to distinguish between the cognitive layers that perform processing and the integrated state that emerges through their interaction. Da'at does not add a new cognitive function; it serves as a concept for describing the integrated state separately from the cognitive layers themselves.

At present, however, it has not been established that treating this distinction as an independent concept called "Da'at" provides explanatory power beyond simply describing it as "the state in which a judgment or understanding has been formed."

Status:

Not an independent cognitive function / a concept used to distinguish the integrated state from the cognitive layers themselves / additional explanatory power remains unconfirmed

Q77. What becomes impossible to describe if Da'at is removed?

Related Qs: Q74–Q76, Q78–Q80

Under the current SCOPE model, it has not been established that removing Da'at would make any cognitive phenomenon itself impossible to describe.

Da'at is not an independent cognitive layer or processing function. It is a concept representing the state in which the cognitive structure has come together as a single judgment or understanding. Therefore, even without the term "Da'at," the same content could still be described as "the state in which the cognitive structure has come together as a single judgment or understanding."

However, removing Da'at would eliminate a concise label for distinguishing the cognitive layers that perform processing from the integrated state in which those processes have come together as a single judgment or understanding. The relationship between this integrated state and output in humans and LLMs would also need to be explained each time rather than referenced through a single concept.

Accordingly, what would clearly be lost at present is mainly the conceptual and descriptive convenience of distinguishing the integrated state concisely. It has not yet been established that there are cognitive phenomena that cannot be described unless Da'at is retained as an independent concept.

Status:

No cognitive phenomenon has been shown to become indescribable / the concise label distinguishing the integrated state would be lost / theoretical indispensability remains unconfirmed

Q78. Can the difference between L8 and Da'at be distinguished observationally?

Related Qs: Q14, Q74–Q80

Within SCOPE, L8 and Da'at are conceptually and structurally distinct. However, at present, no criterion has been established that can independently and uniquely distinguish the two from outputs appearing in Malkuth.

L8 is a cognitive layer that references the cognitive structure as a whole and judges whether that whole can, at that moment, form a single judgment or understanding. The conditions for this judgment are not defined in the current SCOPE framework. The individual paper on Da'at also places the relation in which Da'at emerges when L8 references the cognitive structure as a whole and judges that it can come together as a single judgment or understanding.

Da'at, by contrast, is not a cognitive layer. It is an integration point representing the state in which multiple cognitive structures have been integrated and come together as a single judgment or understanding. Da'at is not itself an independent function that performs cognitive processing.

Accordingly, by definition, L8's judgment of whether formation is possible and Da'at as the resulting state in which cognition has come together as a single judgment or understanding are not the same concept.

However, this conceptual distinction cannot simply be confirmed as an observational distinction.

What can be directly observed in SCOPE is output appearing in Malkuth, such as speech, text, behavior, or generated responses. L8 itself, how L8 references the cognitive structure as a whole, and the point at which it judges that formation is possible cannot be directly observed from the outside. For L8, hypotheses about its involvement must therefore be placed using observable output as a clue.

The integrated state represented by Da'at also cannot be directly observed. When content formed after the emergence of Da'at passes through transmission processing and appears in Malkuth, that output may be used as a clue for examining the possibility that Da'at emerged. However, observable output and the internal integrated state do not correspond one to one.

The individual paper on Da'at also assumes that, in humans, speech, bodily reactions, or other external outputs may appear before cognition has fully come together as a single

judgment or understanding. However, observing such an output alone does not allow the involvement of L8 or the non-emergence of Da'at to be uniquely established.

Observationally, it is therefore possible to distinguish conceptually between a hypothesis that "whole-structure reference and a judgment of possible formation corresponding to L8 were involved" and a hypothesis that "the cognitive structure was integrated into a single judgment or understanding and Da'at emerged." However, at present, no operational criterion has been defined for independently separating, from the same Malkuth output, how far L8 was involved or at what point Da'at emerged.

Therefore, although L8 and Da'at are distinct by definition and structure, they cannot be independently established from observable output alone. When they cannot be distinguished, neither should be uniquely confirmed.

Status: L8 and Da'at are conceptually and structurally distinct / L8 is a cognitive layer that references the cognitive structure as a whole and judges whether it can form a single judgment or understanding, while the conditions for that judgment remain undefined / Da'at is the state or integration point at which cognition has come together as a single judgment or understanding and is not an independent cognitive function / Neither can be directly observed / No operational criterion has been established for uniquely and independently distinguishing them from Malkuth output / When they cannot be distinguished, neither is uniquely confirmed

Q79. If the Criteria for L8's Judgment of Possible Formation Are Undefined, Does Saying "L8 Passed It / Did Not Pass It" Amount to Reasoning Backward from the Output?

Related Qs: Q14, Q78, Q80

If the output alone is used to establish that "L8 passed it" or "L8 did not pass it," then this does amount to reasoning backward from the result. However, inferring cognitive structure in the reverse direction from observable output and placing a hypothesis about the involvement of L8 is itself consistent with the method of SCOPE.

SCOPE is a framework that uses observable output appearing in Malkuth as a clue for placing hypotheses about the cognitive structure behind that output. Therefore, placing a hypothesis about the involvement of L8 on the basis of output is itself permissible. SCOPE does not directly observe the internal layers themselves; it infers the cognitive structure behind observable output.

The issue is whether the judgment criteria used for that inference are sufficiently defined.

In current SCOPE, L8 is positioned as a cognitive layer that references the cognitive structure as a whole and judges whether that whole can, at that moment, form a single judgment or understanding. However, no independent criterion has yet been defined for what L8 uses to judge that formation is possible. The individual paper on Da'at likewise places the

relation in which Da'at emerges when L8 references the cognitive structure as a whole and judges that it can come together, but the criteria for that judgment are themselves undefined.

Accordingly, it is possible to place a hypothesis that, for a given output, "L8 judged formation to be possible." However, the mere fact that the output occurred cannot be used to establish the content of L8's judgment as an observed fact.

In particular, if one infers only that "the output occurred, therefore L8 passed it," or "the expected output did not appear, therefore L8 did not pass it," then the observation itself is being used as the basis for the L8 judgment. Without an independent judgment criterion, this becomes a post hoc explanation.

The relevant question is whether the conditions for L8's judgment can be applied consistently as an inferential criterion without being changed after the observation is known. At present, no such independent condition has been defined. Therefore, whether "L8 passed it" or "L8 did not pass it" cannot be established from an independent criterion.

This does not prohibit placing hypotheses about L8 from observable output. What is restricted is treating the observed result alone as sufficient to establish the content of L8's judgment and then using that established judgment again as an explanation of the same result.

Status: Reverse-direction inference from output is part of SCOPE's method / L8 is assigned the roles of whole-structure reference and judgment of possible formation, but the criteria for that judgment remain undefined / Establishing "L8 passed it / did not pass it" from the output alone amounts to reasoning backward from the result / If an L8 judgment is established solely from an observed result and then used to explain that same result, the explanation is circular / Placing a hypothesis about L8 involvement from output is not itself prohibited / At present, L8's judgment cannot be established from an independent criterion

Q80. If any output can be explained by saying that "L8 judged it to be viable," does that make L8 unfalsifiable?

Related Qs: Q47–Q50, Q74, Q78, Q79, Q88–Q93

If L8 is used in that way, then it becomes unfalsifiable.

In SCOPE, L8 is positioned as the layer that references the cognitive structure as a whole and judges whether that structure can, at that moment, be regarded as viable as a whole. However, at present, no independent criteria have been defined for what L8 uses to make that judgment of viability.

Therefore, if an output is observed and one then explains, "Because the output occurred, L8 must have judged it to be viable," and further uses that judgment to explain why the same output occurred, the circularity discussed in Q79 arises.

Under this use, any observed output could be explained after the fact by saying that “L8 judged it to be viable.” In that case, there would be no condition under which the claim that “L8 judged it to be viable” could be rejected, and L8 would function as an all-purpose post hoc explanation adjusted to the result.

Connecting COS does not automatically supply the missing criteria for L8’s viability judgment. If an undefined judgment is inferred from an output and then used to explain that same output, connecting COS does not by itself make that use falsifiable.

However, this does not mean that the L8 hypothesis itself is immediately unfalsifiable in its entirety. The problem is specifically the use of L8 as a definitive explanation for individual outputs while the criteria for its viability judgment remain undefined.

Therefore, at present, SCOPE assigns L8 the role of referencing the cognitive structure as a whole and judging whether it can be viable, but it is not possible to determine from an individual output alone that “L8 judged it to be viable.” If such a determination is then used to explain the result itself, the explanation becomes unfalsifiable.

Status:

L8 has an existing role definition / criteria for its viability judgment remain undefined / inferring L8’s judgment from an output and then using that judgment to explain the same output creates circularity / applying “L8 allowed it” to any output makes the explanation unfalsifiable / connecting COS does not automatically supply L8’s judgment criteria / L8’s judgment cannot be definitively determined from an individual output

Q81. What must match for a “functional correspondence” to be established in SCOPE?

Related Qs: Q82–Q87, Q88–Q93

In SCOPE, “functional correspondence” does not mean that humans and LLMs share the same internal structure or implementation. Even when their implementations or forms differ, a correspondence may be proposed when the roles or functions selected for comparison show meaningful commonality.

Accordingly, what is compared is not a specific internal component or algorithm, but the role defined within SCOPE.

For example, human L6 concerns the function of constructing reasons or explanatory models about an object. If a similar function is observed in LLM output, it may be treated as a candidate functional correspondence with L6. This does not imply that an independent mechanism called “L6” exists inside the LLM. Likewise, functional labels assigned on the LLM side are comparison labels, not claims about actual technical components.

However, similar output alone is not sufficient to establish functional correspondence, because the same output may arise from different structures or processes. Q82 examines

how much commonality is required, while Q83 examines how far different implementations can still be treated as functionally corresponding.

Status:

The comparison concerns roles or functions rather than internal implementation / LLM-side functional labels do not denote actual technical components / output similarity alone is insufficient / detailed correspondence criteria are examined in Q82 and Q83

Q82. Is similar output alone sufficient to conclude that the same function is involved?

Related Qs: Q81, Q83–Q87

No. Similar output alone is not sufficient to conclude that the same function is involved.

SCOPE uses observable output as a clue for inferring cognitive structure. However, identical or similar outputs may arise from different internal structures or processes. Output similarity can therefore serve as evidence for considering a functional correspondence, but it is not sufficient to establish that correspondence conclusively.

For example, even if both a human and an LLM produce explanations containing similar reasoning, that similarity alone does not show that the same internal structure or process was involved. SCOPE can only treat the outputs as candidates for correspondence with the same functional role.

Accordingly, when observable output cannot distinguish among multiple functional hypotheses, SCOPE does not force a single conclusion. The possibilities are retained as multiple candidates or treated as observationally indistinguishable.

Q83 further examines how much commonality is required before different systems can reasonably be treated as functionally corresponding.

Status:

Output similarity alone is insufficient to establish the same function / similarity can provide evidence for considering functional correspondence / internal structural or process identity cannot be established from output alone / unresolved cases remain as multiple candidates or observationally indistinguishable

Q83. Where is the boundary for calling two systems “the same function” even when their implementations differ?

Related Qs: Q81, Q82, Q84–Q87

In SCOPE, whether two cases are treated as having the “same function” is determined not by whether the objects themselves are identical, but by which role or function has been selected as the basis of comparison.

For example, a smoke signal and a smartphone differ in implementation, medium, speed, bidirectionality, and many other respects. However, if the function being compared is “transmitting information to someone at a distance,” they can be treated as performing the same function from that perspective. If communication method, responsiveness, technological system, or another role is being compared, then that constitutes a different comparison.

Even between the same objects, the judgment may change when the comparison axis changes. For example, a dog and a whale belong to the same category when the axis is “mammal,” but they differ when the axis is “land-dwelling organism.” This is not a contradiction; the compared role or attribute is different.

SCOPE L1 and the COS C-series likewise both deal with language in a broad sense. However, L1 concerns the role of converting cognitive content into a form that can be verbalized, whereas the C-series concerns the process of how that content is communicated. Therefore, when the comparison axis is role, they are treated as different functions.

Thus, differences in implementation do not by themselves invalidate functional correspondence. The boundary lies in whether the role defined in advance for the comparison is shared.

Names assigned to the LLM side, such as “Verbalization Interface” and “Structural Model Formation,” are explanatory labels for roles rather than names of actual technical components. Therefore, the boundary of a function is determined not by the label itself, but by the role definition associated with that label at the time of comparison.

If a counterexample or mismatch appears, the original correspondence cannot be preserved by changing the comparison axis or redefining the meaning of the label afterward. If a different comparison axis is introduced, it must be treated as a separate comparison. If the role definition associated with a label is changed, this must be treated as a new correspondence hypothesis or a theory update rather than as a retroactive reinterpretation of the original definition.

At present, the boundary of “the same function” in SCOPE is therefore not identity of implementation, but whether the role defined in advance as the object of comparison is shared.

Status:

“Whether two cases have the same function is judged by the role defined for comparison, not by implementation.”

“The judgment may change when the comparison axis changes, even for the same objects.”

“Different implementations may still be candidates for functional correspondence when they perform the same role.”

“Function names are labels; the boundary is determined by the associated role definition.”
“If the comparison axis or role definition is changed afterward, it must be treated as a separate comparison or theory update.”

Q84. What happens to SCOPE as a whole if a layer holds for humans but has no corresponding function in LLMs?

Related Qs: Q81–Q83, Q85–Q87

SCOPE does not assume that humans and LLMs have identical internal structures or implementations. What is being compared is whether functional correspondence holds with respect to a role defined in advance.

Therefore, differences in implementation, or the absence in LLMs of psychological or biological states found in humans, do not by themselves falsify functional correspondence. Likewise, the fact that a corresponding function does not appear in observable output, or that the internal state cannot be directly observed, is not sufficient to establish its absence. If the possibilities cannot be distinguished, the case is treated as observationally indistinguishable.

If functional correspondence can be identified as not holding with respect to the predefined role, the human–LLM correspondence hypothesis for that layer is falsified. That layer also cannot continue to be placed unchanged as a shared coordinate for humans and LLMs.

However, this does not mean that the human-side definition of the layer, or SCOPE as a whole, is immediately falsified. In addition, if the compared role is broadened only after a counterexample appears in order to preserve correspondence by claiming that the two are “similar in a broader sense,” this must be treated not as preservation of the original hypothesis, but as a new correspondence hypothesis or a theoretical revision.

Whether the common coordinate system itself can still be maintained when some layers do not correspond is addressed in Q85.

Status:

“Differences in implementation or the absence of human-specific states do not by themselves falsify functional correspondence.”

“The absence of a function is not established merely because it does not appear in output or because internal states cannot be directly observed; if the possibilities cannot be distinguished, the case remains observationally indistinguishable.”

“If non-correspondence can be identified under the predefined role, the human–LLM correspondence hypothesis for that layer is falsified.”

“For a layer identified as non-corresponding, its shared placement for humans and LLMs cannot be maintained unchanged.”

“Falsification of the correspondence hypothesis is distinguished from falsification of the human-side layer definition or SCOPE as a whole.”

“If the comparison axis is broadened after a counterexample, this is treated as a new correspondence hypothesis or a theoretical revision.”

“Whether the common coordinate system itself can still be maintained is addressed in Q85.”

Q85. Can SCOPE Still Be Maintained as a Common Coordinate System If Only Some Layers Do Not Correspond Between Humans and LLMs?

Related Qs: Q81–Q84, Q86–Q87

If, for a particular layer, functional correspondence between humans and LLMs can be identified as failing with respect to the role defined in advance, that layer can no longer be treated as a functional coordinate common to both humans and LLMs.

However, this does not by itself invalidate functional correspondence in the other layers. Other layers may still be used as common coordinates for comparing humans and LLMs, but only where correspondence in those layers can itself be identified. If correspondence cannot be distinguished as either present or absent, that layer cannot be confirmed as remaining part of the common coordinate system.

In that case, the claim that all layers correspond as common coordinates between humans and LLMs can no longer be maintained. SCOPE would require a theoretical update that revises the range of correspondence so that it functions as a coordinate system for comparing humans and LLMs only within the range where correspondence can be identified.

This constitutes a refutation of the human–LLM correspondence hypothesis for part of SCOPE. It does not, by itself, mean that SCOPE as a model of human cognition, or other layers whose correspondence can still be identified, have also been refuted.

How to treat cases in which the same coordinate is considered functionally corresponding but produces different behavior in humans and LLMs is addressed in Q86.

Status:

“If functional correspondence is identified as failing for a particular layer, that layer cannot be maintained as a functional coordinate common to humans and LLMs.”

“Failure of correspondence in one layer does not automatically refute correspondence in the other layers.”

“Other layers can remain common coordinates only where their correspondence can be identified.”

“If partial non-correspondence is identified, the claim that all layers are common can no longer be maintained, and a theoretical update revising the range of correspondence is required.”

“Refutation of correspondence for part of SCOPE must be distinguished from refutation of SCOPE as a model of human cognition.”

“Cases in which the same coordinate produces different behavior in humans and LLMs are addressed in Q86.”

Q86. How Should the Correspondence Be Treated When Humans and LLMs Are Considered to Correspond to the Same Coordinate but Different Behaviors Are Observed?

Related Qs: Q81–Q85, Q87

In SCOPE, even when a correspondence hypothesis is placed between humans and LLMs at the same common coordinate, this does not require the same input to produce the same output. What SCOPE compares is whether functional correspondence exists with respect to a role defined in advance, not whether internal structures or output results themselves are identical.

Therefore, observing different behavior in humans and LLMs does not by itself refute functional correspondence. At the same time, every observed difference cannot simply be attributed to different conditions in order to preserve the correspondence.

If the observed difference makes it possible to identify that the role defined in advance is not the same on the human side and the LLM side, the functional correspondence hypothesis for that coordinate is refuted. If it cannot be distinguished whether the difference reflects different manifestations of the same role or a mismatch in the role itself, the correspondence is not confirmed as either established or failed and is treated as indeterminate.

Accordingly, the issue in Q86 is not whether the outputs are the same or different, but whether it is possible to distinguish between a difference that is compatible with the role defined in advance and a difference that indicates a mismatch in the role itself.

Whether humans and LLMs can be regarded as having the same manner of operation or the same dynamics even when they correspond to the same coordinate is addressed in Q87.

Status:

“A correspondence hypothesis at the same common coordinate does not require the same input to produce the same output.”

“Different observed behavior alone does not refute functional correspondence.”

“Different behavior cannot be unconditionally attributed to different conditions in order to preserve the correspondence.”

“If it can be identified that the role defined in advance is not the same on the human side and the LLM side, the correspondence hypothesis for that coordinate is refuted.”

“If it cannot be distinguished whether the difference is a different manifestation of the same role or a mismatch in the role itself, it is treated as indeterminate.”

“The same coordinate and the same manner of operation or dynamics are not treated as identical. Details are addressed in Q87.”

Q87. Does the Ability to Compare Humans and LLMs Using the Same Coordinate Mean That They Have the Same Internal Structure or Operate According to the Same Dynamics?

Related Qs: Q81–Q86

No.

SCOPE does not assume that humans and LLMs have identical internal structures or implementations. What is compared using common coordinates is whether a hypothesis of functional correspondence can be placed with respect to a role defined in advance.

Therefore, even when humans and LLMs can be compared using the same coordinate, this does not mean that the role is formed, changes, or appears in output in the same way internally.

Here, “the same coordinate” means that the two can be compared in terms of the same kind of role. By contrast, “the same dynamics” means that even the way the role changes and interacts is shared. SCOPE treats the former as a hypothesis, but does not assume the latter.

Even when a functional correspondence hypothesis can be placed between humans and LLMs at the same coordinate, differences in internal structure or rules of change can still coexist with that correspondence. Conversely, the fact that humans and LLMs can be placed at the same coordinate does not by itself justify inferring that they operate through the same mechanism.

Common coordinates in SCOPE are not intended to treat humans and LLMs as the same internal model. They are coordinates for comparing and describing corresponding roles using a common reference, even across entities that may have different internal structures.

Status:

“Correspondence at the same coordinate does not imply the same internal structure or implementation.”

“Correspondence at the same coordinate does not imply the same rules of change or interaction.”

“What SCOPE makes common is the coordinate used to compare roles defined in advance.”

“SCOPE does not claim that humans and LLMs operate through the same mechanism.”

“Even entities that may have different internal structures can be compared and described using common coordinates where a functional correspondence hypothesis can be placed.”

Q88. When the same cognitive phenomenon is examined, how are the roles of SCOPE and COS distinguished?

Related Qs: Q80, Q87, Q89–Q93

Revision history: This question incorporates the former wording, “Where does SCOPE end and COS begin?” The issue of whether a boundary exists is retained as part of the distinction, without assuming that there is a single dividing line between the two.

SCOPE and COS may use the same cognitive phenomenon as material for analysis, but they are not two models that describe the same thing under different names.

SCOPE is a coordinate system for provisionally placing, comparing, and inferring cognitive structures behind observable speech, text, behavior, generated responses, and other outputs. COS, by contrast, is a processing model for organizing what kinds of cognitive processing are occurring, such as decision-making, interpretation, communication, and observation.

The same output can therefore be examined from both perspectives. SCOPE considers what kind of cognitive structure may provisionally be placed behind the output, while COS organizes the cognitive processes related to that output.

Accordingly, there is no single dividing line at which “SCOPE ends and COS begins.” The two are not used by moving from one to the other in temporal sequence. Rather, for the same cognitive phenomenon, SCOPE primarily describes cognitive structure, whereas COS primarily describes cognitive processing.

Possible correspondences between the two can be examined as hypotheses, but no fixed correspondence table has been established. Nor does the use of COS automatically fill undefined areas within SCOPE.

At present, this division of roles exists at the level of definition, while no prior criterion has been specified for observationally assigning the respective roles of SCOPE and COS from a single output alone. Therefore, an inability to uniquely identify this assignment from observation does not itself negate the distinction between their roles. However, SCOPE does not claim that a dividing line between the two has been observationally confirmed, nor that both models are always necessary for every cognitive phenomenon.

Status:

“SCOPE and COS may address the same cognitive phenomenon, but they are not two models that describe the same thing under different names.”

“SCOPE primarily describes cognitive structure, whereas COS primarily describes cognitive processing.”

“SCOPE does not assume a single dividing line in which processing moves from SCOPE to COS in temporal sequence.”

“Correspondences between SCOPE and COS may be examined as hypotheses, but no fixed correspondence table has been established.”

“Using COS does not automatically fill undefined areas within SCOPE.”

“At present, no prior criterion has been specified for uniquely assigning the respective roles of SCOPE and COS from a single output alone.”

“An inability to uniquely distinguish their roles observationally does not itself negate the role distinction, but SCOPE does not claim that a dividing line has been observationally confirmed or that both models are always required.”

Q89. Can a single cognitive structure really be separated from updating over time?

Related Qs: Q57, Q72, Q80, Q88, Q90–Q92

The estimation of a cognitive structure at a given point and updating over time cannot be separated simply by the length of the time interval. They can, however, be distinguished by what they describe. At present, no prior criterion has been specified for uniquely identifying from observation that the latter occurred as an extension of the former.

SCOPE uses the output observed at each point in time as a basis for provisionally placing a cognitive structure for that point. COS addresses processes such as decision-making, interpretation, communication, and observation, as well as the organization of changes in those processes. The mere fact that multiple outputs occur in temporal sequence is not evidence that the cognitive structure has been updated.

Even when a later output appears externally to contradict an earlier one, a hypothesis in which the target is divided differently or different premises apply may also be consistent with a hypothesis in which the same self-consistency is maintained. In such a case, the passage of time does not necessarily mean that the structure has changed. Conversely, even when outputs differ across multiple points in time, the output difference alone cannot determine whether the cause is a structural change, a difference in target, a difference in premises, or an update on the processing side.

Therefore, even across multiple points in time, the provisional placement of the structure at each point remains within SCOPE. How observed discrepancies or differences are handled as continuing processes can be organized through COS. This is not a handoff in which “a single point in time belongs to SCOPE and multiple points in time belong to COS.” The same sequence of observations can be described along both a structural axis and a processing axis.

Accordingly, a structural update cannot be identified from temporal differences or output differences alone. At present, what can be maintained is the division of roles in which structural estimation at each point and processing or updating over time are treated as different descriptive targets. A criterion for identifying whether an update actually occurred has not yet been established. The inability to make that identification does not negate this division of roles.

Status:

“Time length is not the basis of the distinction; the descriptive target is.”

“A temporal sequence of outputs is not the same as a structural update.”

“Even across multiple points in time, provisional structural placement at each point remains within SCOPE. There is no handoff of ‘single point = SCOPE / multiple points = COS.’”

“A hypothesis involving differences in target or premises may be consistent with a hypothesis in which the same self-consistency is maintained.”

“The cause of an output difference—structural change, target difference, premise difference, or processing-side update—cannot be determined from the output difference alone.”

“No prior criterion has been specified for uniquely identifying update versus non-update. Inability to identify this does not negate the division of roles.”

Q90. When the Term “Update” Is Used Within SCOPE, How Is It Distinguished from Temporal Updating in COS?

Related Qs: Q72, Q88, Q89, Q91–Q93

Both SCOPE and COS may use the term “update,” but the use of the same word does not necessarily mean that the same phenomenon or the same operation is being described. To distinguish them, it is necessary to specify whose state changed and what changed.

In descriptions involving SCOPE, it is necessary to distinguish between a case in which the cognitive structure of the target itself is inferred to have changed and a case in which an analyst using SCOPE revises an inferential hypothesis previously placed about that cognitive structure in response to new observations. The latter does not mean that the target person’s cognitive structure itself has changed.

COS, by contrast, addresses cognitive processing. Within COS, premises, interpretations, hypotheses, and other elements being processed may be revised in response to new information. In addition, a model being maintained about a target may be continuously updated across multiple observations. These are treated as updates in processing, distinct from changes in the cognitive structure itself.

Accordingly, the word “update” alone does not determine whether the target’s cognitive structure changed, whether the analyst’s inferential hypothesis was revised, or whether processing within COS was revised over time. In description, these should not be treated as identical, and whose state changed and what changed should be made explicit.

However, it may not always be possible to distinguish these uniquely from observable output alone. If it cannot be identified which type of change should be assigned, it should not be forced into any one category.

The continued operation of a target model across multiple observations is not treated as a function of SCOPE alone, but as an area of connection with COS. SCOPE is a coordinate system that infers cognitive structure using output observed at a given point as a clue; it is not itself treated as the mechanism responsible for continuous model operation over time.

Status:

“The term ‘update’ is not treated as referring to the same phenomenon or the same operation in SCOPE and COS.”

“A change in the target’s cognitive structure is distinguished from a revision of an inferential hypothesis by the analyst.”

“A revision of the analyst’s inferential hypothesis does not mean that the target person’s cognitive structure itself has changed.”

“Within COS, premises, interpretations, hypotheses, and other elements being processed may be revised in response to new information.”

“When describing an ‘update,’ it is necessary to specify whose state changed and what changed.”

“If the type of update cannot be uniquely identified from observation, it should not be forced into a definite category.”

“No prior criterion has currently been specified for assigning an observed change uniquely among a change in the target’s cognitive structure, a revision of the analyst’s inference, and an update in COS processing.”

“Continuous operation of a target model across multiple observations is treated as an area of connection with COS rather than as a function of SCOPE alone.”

Q91. Does Using Multiple Observations Automatically Mean That the Case Should Be Described in Terms of COS?

Related Qs: Q88–Q90, Q92–Q93

Not automatically.

SCOPE and COS are not distinguished by the number of observations or by the length of time involved. They are distinguished by the roles of the models and by what each model is intended to describe.

SCOPE is a coordinate system used to place, compare, and examine hypotheses about underlying cognitive structure on the basis of observable output. L0–L8 do not represent stages of processing; they represent positions used to organize and infer cognitive structure. Therefore, the mere fact that multiple observations are used does not determine that the case should be described as COS processing.

COS is a processing structure that addresses cognitive processes such as decision-making, interpretation, transmission, and observation. However, COS does not guarantee correct judgment or correct interpretation.

Even when multiple observations are related across time, the description may still be made in terms of SCOPE if those observations are being compared in order to examine what positions or structural relationships may be involved in the cognitive structure. By contrast, when retained hypotheses or a target model are being tested or updated as part of an ongoing cognitive process, the description may be made in terms of COS. Therefore, the number of observations or the fact that they are arranged over time is not sufficient by itself to assign the case uniquely to either model.

At present, no prior criterion has been specified for uniquely assigning an observed case to either a SCOPE role or a COS role. When the distinction cannot be made, the case should not be forced into either one.

The same utterance or the same observational target may also be described from both SCOPE and COS perspectives. For example, a state in which cognitive structures have come together as a single judgment or understanding may be described as Da’at, while the

processing that adjusts how that content is conveyed may be described in terms of the C series of COS. This does not mean that SCOPE itself performs integration or transmission. Nor should this relationship be treated as a fixed processing sequence.

Accordingly, SCOPE and COS are not distinguished by a rule such as “one observation means SCOPE, multiple observations mean COS.” Multiple observations are not a sufficient condition for describing a case in terms of COS, and COS processing may also be involved in a single observation. SCOPE and COS are distinguished not as regions that a case moves between, but as models that describe the same target from different functional roles.

Status:

“Multiple observations alone do not determine that a case should be described in terms of COS.”

“SCOPE and COS are not distinguished by the number of observations or by the length of time involved.”

“SCOPE is a coordinate system for placing, comparing, and examining hypotheses about cognitive structure.”

“L0–L8 are not processing stages but positions used to organize and infer cognitive structure.”

“COS addresses cognitive processes including decision-making, interpretation, transmission, and observation.”

“COS does not guarantee correct judgment or correct interpretation.”

“Even when multiple observations are used across time, that fact alone does not determine that the case should be described in terms of COS.”

“No prior criterion has currently been specified for uniquely assigning an observed case to a SCOPE role or a COS role.”

“When the distinction cannot be made, the case should not be forced into either one.”

“The same utterance or observational target may be described from both SCOPE and COS perspectives.”

“Da’at is an integrated state; SCOPE itself is not treated as an integration mechanism.”

“SCOPE and COS are not treated as a fixed before-and-after sequence or as regions between which a case moves.”

Q92. What is the difference between “repositioning a hypothesis” within SCOPE and “updating a target model” within COS?

Related Qs: Q88–Q91, Q93

The two are similar in that a new observation may change a previous view. However, what is being changed is different.

In SCOPE, the “repositioning of hypotheses” means that an analyst reexamines the inferential hypotheses they had placed about a target’s cognitive structure, using observable outputs as evidence.

What directly changes in the repositioning of hypotheses within SCOPE is the inferential hypothesis held by the analyst. This alone does not mean that the target person's cognitive structure has changed. The possibility that the target person's cognitive structure itself has changed is treated as a separate hypothesis from the analyst's revision of the inference.

When a single conclusion cannot be reached, multiple hypotheses may also be retained and compared without forcing them into one conclusion.

Therefore, the fact that a hypothesis has been repositioned within SCOPE does not by itself establish either that the target person's cognitive structure has changed or that a target model in COS has been updated.

By contrast, D4 in COS concerns the process of operating a target model for a specific target, comparing and testing it against actual observations, and updating it when necessary. In D4, the model of understanding that had previously been used for the target may be revised through comparison with observed results.

Hypotheses or the results of structural research organized in D3 may become material for D4, but this does not imply a fixed sequence in which D3 must always lead to D4. The processing systems within COS do not operate in a simple linear order; they may influence one another as needed.

Accordingly, in the repositioning of hypotheses within SCOPE, the inferential hypothesis used analytically to describe a cognitive structure is reconsidered.

In the updating of a target model within COS, D4 describes the process of comparing and testing the target model being used for a specific target against observations and updating it when necessary.

The same new observation may serve, in an analysis using SCOPE, as material for reconsidering an inferential hypothesis about cognitive structure, while in COS it may serve as material for comparing, testing, and updating a target model in D4. This does not mean that the two are the same process.

Likewise, the mere fact that a view changed after the same observation does not determine whether what occurred was a repositioning of a SCOPE hypothesis, a change in the target person's cognitive structure, or an update of a target model in COS.

When an analyst is repositioning an inferential hypothesis about cognitive structure, this can be described as the repositioning of a hypothesis within SCOPE.

When a target model currently being used for a specific target is compared with observations and revised, this can be described as target-model updating within COS.

When there is a possibility that the target person's cognitive structure itself has changed, that possibility is not equated with the analyst's revision of a hypothesis, but is treated as a separate hypothesis concerning the target's cognitive structure.

The two may appear to operate in conjunction, but they do not form a fixed processing sequence. Nor does this mean that the domain shifts from SCOPE to COS, or that the two are the same process.

At present, when a change in the target person's cognitive structure, a repositioning of an inferential hypothesis within SCOPE, and an update of a target model within COS cannot be uniquely distinguished, the analysis does not force the observation into only one of these interpretations.

Status:

"In the repositioning of hypotheses within SCOPE, what directly changes is the analyst's inferential hypothesis about cognitive structure."

"This alone does not mean that the target person's cognitive structure has changed."

"A change in the target person's cognitive structure is treated as a separate hypothesis from the analyst's revision of an inference."

"In COS D4, a target model being used for a specific target is compared and tested against observations and updated when necessary."

"D3 and D4 are not treated as a fixed linear sequence."

"The same observation may provide material for reconsideration in both cases, but the two are not treated as the same process."

"When a change in the target structure, a revision of a SCOPE inference, and an update of a COS target model cannot be distinguished, the analysis does not force the case into only one category."

Q93. If SCOPE and COS have distinct roles but their descriptions overlap in individual cases, is it necessary to separate them into two models?

Related questions: Q74, Q80, Q88–Q92

At present, there is a reason to describe them separately. However, it has not been demonstrated that they must necessarily exist as two models.

SCOPE is a coordinate system for placing inferences about cognitive structure, whereas COS is a processing structure for organizing cognitive processing. Even when the same output can be described from both perspectives, this does not necessarily mean that they are describing the same thing. The reason for separating them is to avoid conflating inferences about cognitive structure with descriptions of cognitive processing.

Being able to preserve this distinction by describing them separately is not the same as demonstrating that two models are necessary. No comparative validation has yet been conducted against an alternative in which the same distinction is maintained within a single model. It has also not been shown that a two-model structure is more advantageous, or that the distinction cannot be preserved without separating the models.

If an integrated model can describe the same phenomena equally well or better while preserving the distinction between structural inference and processing description, the need

to maintain two separate models would become weaker. Conversely, if integrating them makes distinctions that are currently observable or analytically separable no longer distinguishable, this would strengthen the reason for keeping the two models separate.

Overlap in individual cases must not be used to assign a case conveniently to either SCOPE or COS in order to eliminate counterexamples or unresolved identification problems. When the distinction cannot be made, the case should remain classified as unresolved with respect to that distinction.

Status:

“There is a reason to describe SCOPE and COS separately.”

“The necessity of a two-model structure has not been demonstrated.”

“Being describable from both perspectives does not mean the two models are identical.”

“No comparison with an integrated alternative has yet been conducted.”

“Cases that cannot be distinguished should not be forced into either model.”

Q94. What counts as a counterexample to SCOPE?

Related questions: Q3, Q7–Q13, Q51–Q57, Q80, Q83, Q88–Q93, Q95–Q103

A counterexample to SCOPE is not simply a case that is difficult to explain or difficult to classify immediately.

An observation becomes a candidate counterexample when it is incompatible with a specific proposition, distinction, connection, correspondence, or falsification condition that had already been specified in SCOPE at that time.

However, a single observation does not always establish a counterexample immediately. It should first be treated as a candidate counterexample. If the incompatibility remains after checking observational conditions, possible measurement problems, and competing interpretations, it can then be treated as a counterexample to that specific proposition.

If a point remains undefined, failure to explain it is not treated as a counterexample but as an unresolved issue. Likewise, when a proposition already has an explicit defeat condition, an observation satisfying that condition becomes a candidate counterexample. A convenient defeat condition must not be added only after a counterexample appears in order to reverse the result.

Facts such as “the case cannot be classified,” “multiple layers appear highly active at the same time,” or “a unique inference cannot be made” do not automatically constitute counterexamples. However, if SCOPE explicitly claims that the case should be uniquely identifiable under those conditions, failure of identification may itself become a candidate counterexample.

The same applies to slide relationships. In the current version, where activation amount and ratio are distinguished, the observation that “both activation amounts are high” does not by itself refute a slide relationship. The relevant question is whether an observation that violates the proposed relationship still occurs when the target and cognitive unit are held constant.

If, after a counterexample appears, additional explanations involving another layer, another target, activation amount, COS, L8, or similar concepts are introduced and make the case explainable, this does not erase the original counterexample. If the existing theory could not answer the case and an explanation was added in response to it, that change should be recorded as a theoretical update.

Refuting one proposition does not automatically refute SCOPE as a whole. The affected scope should be recorded separately: whether the result requires a local revision, revision of an individual layer, a connection, a core proposition, or SCOPE as a whole. Conversely, the survival of SCOPE as a whole must not be used to erase the failure of an individual proposition.

Keeping a candidate counterexample indefinitely in the status of “not yet confirmed” may also become a form of falsification avoidance. If every possible observation can be made consistent with the existing SCOPE framework and no observation can be specified under which a proposition would fail, the issue should be reconsidered as a problem of unfalsifiability.

Status:

“A counterexample is an observation incompatible with a specific proposition that had already been specified at that time.”

“A case is first examined as a candidate counterexample, and is treated as a counterexample when incompatibility remains.”

“Undefined points remain unresolved rather than being treated as counterexamples.”

“Explanations added after a counterexample are recorded as theoretical updates.”

“Partial falsification is distinguished from falsification of SCOPE as a whole.”

“A framework that can explain every possible outcome should be examined for unfalsifiability.”

Q95. What counts as being outside the scope of SCOPE?

Related questions: Q3, Q7–Q14, Q81–Q87, Q94, Q96–Q103

Being outside the scope of SCOPE refers to questions, phenomena, or levels of explanation that SCOPE had not previously defined as objects it directly addresses.

SCOPE is a cognitive coordinate system for organizing and inferring possible cognitive structures from observable outputs. It does not directly observe the layers themselves.

Therefore, SCOPE does not, on its own, directly establish correspondences between its layers and entities, mechanisms, classifications, diagnoses, scales, ability measures, outcome measures, or internal implementations defined by external academic or evaluative systems. For example, SCOPE layers cannot be mapped one-to-one, without separate validation, onto specific brain regions, neural mechanisms, psychological or medical classifications, personality typologies, measures such as IQ or EQ, occupational aptitude, outcome indicators, or particular internal modules of an LLM.

This does not rule out comparison with external theories or empirical research, nor does it rule out testing correlations or possible correspondences. When such relationships are examined, claims derived from SCOPE's internal definitions should be distinguished from findings obtained through external comparison or validation.

Observable outputs used by SCOPE are not limited to speech. However, the fact that something is observable does not mean that a cognitive structure can be inferred from it with sufficient justification.

When outputs are sparse, their referents or context are unclear, or the relationship between speech and behavior cannot be interpreted uniquely, there may be insufficient evidence to distinguish among competing cognitive hypotheses. In such cases, the target should not automatically be placed outside SCOPE. The case should instead remain indeterminate, observationally indistinguishable, or unresolved.

In the SCOPE source materials reviewed so far, no rule has been identified that excludes a person solely on the basis of age, diagnosis, or disability. These attributes alone therefore cannot establish that a case lies outside the scope of SCOPE.

Nonverbal behavior may also constitute observable output. However, the extent to which cognitive structure can be inferred from particular patterns of gaze, posture, facial expression, or gesture has not yet been sufficiently specified or validated. Nonverbal output itself is therefore not treated as outside the scope of SCOPE, while the validity of inferences drawn from it remains partly untested.

SCOPE also does not uniquely determine a person's communicative intention or internal state from observed speech or behavior. The same output may be compatible with multiple cognitive states or interpretations.

Accordingly, being outside the scope of SCOPE must be distinguished from cases that fall within its observational domain but cannot currently be inferred with sufficient confidence. The former concerns objects or levels of explanation that SCOPE does not directly claim to explain or identify. The latter concerns cases in which the available output does not provide enough evidence to narrow the cognitive hypotheses adequately.

A case must also not be reclassified as "outside the scope of SCOPE" only after a counterexample appears. Cases that had already been explicitly excluded before the counterexample must be distinguished from cases excluded only afterward, and the latter should be recorded as a theoretical update.

Status:

"Outside the scope refers to questions, phenomena, or levels of explanation that SCOPE had not previously defined as objects it directly addresses."

"SCOPE does not directly establish correspondences with external academic systems, classifications, scales, ability measures, outcome measures, or internal implementations on its own."

"Comparison and validation against external theories or empirical research are not ruled out."

“Observable output is not limited to speech.”

“Inability to infer is not the same as being outside the scope.”

“Age, diagnosis, or disability alone does not establish that a case is outside the scope.”

“Nonverbal output is not itself excluded, although the validity of inference from it remains partly untested.”

“SCOPE does not uniquely determine communicative intention or internal state from output alone.”

“Post hoc exclusion after a counterexample is recorded as a theoretical update.”

Q96. Is it permissible to redefine something as “outside the scope” after a counterexample has appeared?

Related Qs: Q94, Q95, Q97–Q103

As a general rule, after a counterexample has appeared, it is not permissible to retroactively treat the earlier version of SCOPE as though the counterexample had “originally been outside the scope” merely in order to avoid the counterexample.

If the relevant object had already been explicitly defined as outside the scope in publicly available materials before the counterexample appeared, it may be treated as part of the pre-existing scope boundary. By contrast, if the scope is changed for the first time after a counterexample has been presented, that change should not be treated as an answer provided by the existing version of SCOPE. It should instead be distinguished as a theory update made in response to the counterexample.

Changing the scope as part of a theory update is not itself prohibited. However, such a change must not be used to retroactively invalidate a counterexample that genuinely applied to the earlier version of SCOPE. It is necessary to record what the scope was before the change, which counterexample led to the revision, and which propositions were modified.

In addition, when a counterexample is presented, changing the scope does not necessarily allow SCOPE to be preserved. Depending on the content of the counterexample, it may be necessary to revise individual propositions, layer structure, connection structure, or core assumptions. If the theory still cannot remain coherent after such revisions, the possibility of rejecting part or all of SCOPE must also be considered.

Therefore, a post hoc scope change cannot be used as a universal escape mechanism for preserving SCOPE. Pre-existing exclusions from scope must be distinguished from scope changes made in response to counterexamples, and the latter must be recorded as theory updates. At the same time, the extent to which the counterexample affects SCOPE must be reconsidered.

Status:

“A post hoc change in scope cannot erase a counterexample that applied to the earlier theory.”

“Pre-existing exclusions from scope must be distinguished from scope changes made after a counterexample.”

“A post-counterexample scope change is recorded as a theory update, not as an answer from the existing theory.”

“A theory update does not necessarily preserve SCOPE; outcomes from partial revision to complete rejection must remain possible.”

Q97. How do we distinguish between something that was already outside SCOPE’s scope before the theory was published and something that was moved outside the scope in response to a counterexample?

Related Qs: Q94–Q96, Q98–Q103

The distinction is made by determining whether the exclusion from scope had already been stated in publicly accessible materials that could be verified by third parties before the counterexample was presented.

In SCOPE, it is not possible for outside observers to directly verify what the author had intended before publication. Therefore, the author’s internal intention alone — such as “I had always intended it that way” or “I originally considered it outside the scope” — cannot be used to retroactively establish an earlier scope boundary.

The determination should instead rely on publicly verifiable records that establish a timeline, such as publication dates, version numbers, revision histories, and the text that was publicly available at the relevant time. A version number itself is not required; what matters is whether it is possible to determine what content was publicly available at a given point in time.

If an object had already been explicitly identified as outside the scope in public materials before the counterexample was presented, it is treated as a pre-existing exclusion from scope. By contrast, if no such exclusion appeared in the publicly available materials before the counterexample and the exclusion was introduced only afterward, it is treated as a scope change made in response to the counterexample.

Private conversations, internal notes, unpublished drafts, and retrospective explanations by the author are not treated as public records establishing the scope before the counterexample.

When identical content has been published through multiple media, the relevant publication point is not determined by the platform itself, but by the earliest point at which the content became publicly accessible to third parties. If the content differs across media, the materials are not treated as the same version, and their respective publication times and contents are examined separately.

If multiple public materials published before the counterexample contain inconsistent scope definitions, one of them should not be selected retroactively as the “correct” version. Instead, the scope at that time should be recorded as inconsistent or unclear.

If the relevant public records from before the counterexample cannot be verified and the historical scope cannot be reconstructed, the case should not be classified either as a pre-existing exclusion or as a post-counterexample scope change. It should instead be treated as indeterminate.

Therefore, whether something was “originally outside the scope” or was moved outside the scope after a counterexample is determined not by the author’s internal intention or retrospective explanation, but by publicly available records that existed before the counterexample and by their chronological relationship.

Status:

“Pre-existing exclusions from scope are distinguished from exclusions introduced after a counterexample by examining public records that existed beforehand.”

“The author’s internal intention or retrospective explanation alone cannot retroactively establish an earlier scope boundary.”

“Publicly verifiable records, such as publication dates, version numbers, revision histories, and the text available at the time, are used to establish the timeline.”

“When identical content appears across multiple media, the earliest point at which it became publicly accessible is treated as the initial publication point.”

“If pre-counterexample public materials contain inconsistent scope definitions, the scope at that time is treated as inconsistent or unclear.”

“If the relevant pre-counterexample public record cannot be verified, the case is treated as indeterminate.”

Q98. If auxiliary hypotheses are added every time a new counterexample appears, does SCOPE become unfalsifiable?

Related Qs: Q94–Q97, Q99–Q103

That possibility exists.

Adding an auxiliary hypothesis is not, by itself, necessarily problematic as a theory update. New observations or counterexamples may reveal conditions or distinctions that the existing theory did not adequately address, making the addition or revision of hypotheses necessary.

However, if an auxiliary hypothesis is added each time a new counterexample appears solely to avoid that counterexample, SCOPE may reach a state in which any observation can be explained as consistent with the theory. In that case, SCOPE would risk losing the possibility of being weakened by counterevidence and becoming unfalsifiable.

For this reason, when an auxiliary hypothesis is introduced, it should first be made explicit under what conditions the hypothesis applies and under what conditions it does not. The extent to which verification conditions and falsification conditions have been established should then be treated separately. Here, “verification conditions” refers to observational conditions that could provide evidence in support of the hypothesis.

When both verification conditions and falsification conditions can be specified, they should be stated explicitly so that observations that provide support for the hypothesis can be distinguished from observations that would weaken it, require revision, or lead to rejection.

On the other hand, even if the verification conditions by which a hypothesis could be supported have not yet been established, it may still be possible to specify a falsification condition under which a particular observation would show that the hypothesis does not hold. In such a case, the lack of established verification conditions must not be used to invalidate the falsification condition.

Conversely, even if verification conditions exist, falsifiability remains problematic if the hypothesis can be maintained regardless of what observations occur. If neither verification conditions nor falsification conditions have yet been established, the hypothesis should be explicitly treated as unverified and should not be treated as a supported hypothesis.

The fact that verification or falsification conditions have not yet been established does not, by itself, require the immediate rejection of a hypothesis. However, undefined concepts or conditions must not be used whenever necessary as explanations to avoid counterexamples. If a hypothesis for which it is unclear what observations would support or weaken it is used only when a counterexample appears, it risks becoming unfalsifiable.

An auxiliary hypothesis must also not be used to retroactively erase a counterexample that genuinely applied to the earlier version of SCOPE. In such a case, the change should be recorded not as an answer already contained in the existing theory, but as a theory update made in response to the counterexample.

Furthermore, if explanations are shifted after the fact to “another layer,” “another target,” “activation level,” “COS,” “L8,” or similar categories so that every observation can be described as consistent with SCOPE, this should not be treated as increased explanatory power. It should instead be reconsidered as a possible loss of falsifiability.

Therefore, auxiliary hypotheses may be added, but the presence or absence of verification and falsification conditions should be distinguished, and unresolved conditions should be recorded as unresolved. Auxiliary hypotheses must not turn SCOPE into a structure that cannot be weakened, revised, or rejected by any possible observation.

Status:

“Adding auxiliary hypotheses is not prohibited in itself.”

“Auxiliary hypotheses should not be added without limit solely to avoid counterexamples.”

“The conditions under which an auxiliary hypothesis applies should be stated explicitly.”

“Verification conditions and falsification conditions should be recorded separately.”

“Even when verification conditions are not yet established, any available falsification conditions should be preserved.”

“If neither verification nor falsification conditions have been established, the hypothesis should be treated as unverified.”

“Undefined concepts should not be used only when counterexamples appear as escape mechanisms.”

“Adding an auxiliary hypothesis does not erase counterexamples that applied to the earlier theory.”

“Post hoc explanations should not make every possible observation consistent with SCOPE.”

Q99. How can we determine whether explanations such as “a different layer,” “a different object,” “activation level,” “the COS domain,” or “multiple hypotheses” constitute legitimate scope clarification or avoidance of a counterexample?

Related Qs: Q94–Q98, Q100–Q103

Explanations such as “a different layer,” “a different object,” “activation level,” “the COS domain,” or “multiple hypotheses” are not necessarily forms of counterexample avoidance in themselves. Some of these distinctions and methods already exist within SCOPE or related theories independently of the counterexample in question.

However, the fact that an explanation already appears in existing materials is not sufficient to establish that its use is legitimate. Even an existing distinction may function as a mechanism for absorbing counterexamples if its conditions of application are unclear and it can be applied retrospectively to virtually any counterexample.

The procedure described in this section does not introduce a new cognitive mechanism or a new layer definition into SCOPE. It is an audit procedure for examining whether an existing distinction or explanation is being used to organize a counterexample legitimately or merely to avoid it.

First, it is necessary to determine whether the distinction itself existed as part of the existing theory before the relevant counterexample appeared. If the distinction already existed, it may be treated as a candidate explanation within the existing theory. If it was introduced only after the counterexample appeared, it should not be treated as an answer already contained in the existing theory, but as a candidate theoretical update made in response to the counterexample.

However, prior existence does not mean that the explanation is correct. What matters is whether the definition, unit of analysis, boundary, conditions, or other relevant criteria have been changed in response to the result of the counterexample in order to make the existing distinction fit the case.

For example, when explaining a case as involving “a different object,” it is necessary to examine what counts as the same object and what counts as a different object. If the object is subdivided or the unit of analysis is changed only after a counterexample appears in order to avoid it, the explanation should not be treated as an answer already contained in the existing theory. It should instead be recorded as a theoretical update made in response to the counterexample. If that change is made solely to avoid the counterexample and has no

conditions of application or defeat conditions, it should be re-examined as possible counterexample avoidance.

The same applies to explanations involving “a different layer.” The fact that multiple layer hypotheses may be consistent with an observed output does not itself conflict with SCOPE’s method of inference. However, when an observation is unfavorable to one layer hypothesis, the explanation must not simply be shifted to another layer repeatedly in such a way that no layer hypothesis can ever be weakened or excluded.

It is also necessary to examine how far such distinctions can be identified from observable information. Even if different layers, different objects, or differences in activation level can be proposed in theory, if there are no observable clues that allow them to be distinguished, the explanation should not be fixed as a single conclusion. It should instead be treated as observationally indeterminate. When multiple explanatory hypotheses genuinely remain viable, they may be retained in parallel.

The fact that the conditions of application or identification have not yet been sufficiently established does not require the concept itself to be immediately rejected. However, an underdeveloped concept must not be used as a conclusive explanation only when a counterexample appears in order to invalidate that counterexample.

The same applies to activation level. The fact that activation level already exists as a concept is not sufficient. If the observations that should count as differences in activation level have not yet been established, activation level may be retained as a candidate explanation, but it cannot be used conclusively to say, “This is not a counterexample because the activation level was different.” If virtually any observation can be explained as a difference in activation level, then activation level itself must be reconsidered as a possible auxiliary variable that absorbs counterexamples.

An explanation referring to the “COS domain” is also not necessarily a form of counterexample avoidance. SCOPE is a coordinate system for organizing and inferring cognitive structure, whereas COS is a model for processing such as decision-making, interpretation, communication, and observation. This difference in role is an existing distinction.

Accordingly, when the issue concerns not the inference of cognitive structure or layer correspondence, but the processing performed on the basis of that structure—such as decision-making, interpretation, communication, observation, or the time-directed updating of an object model—“the COS domain” may be used as a candidate explanation based on the existing division of roles.

However, the existence of that role distinction alone is not sufficient. When an observation is incompatible with an individual SCOPE proposition or structural hypothesis, that observation must not be moved retrospectively to the COS side in order to invalidate the counterexample. The classification is legitimate only when the pre-existing role definitions can be applied without changing their meaning or the unit of analysis in response to the counterexample. If the meaning of those roles must be changed in order to apply the

distinction, the change should be treated as a theoretical update rather than as a straightforward application of the existing distinction.

When a single phenomenon includes both structural inference and processing, it should not be forced entirely into one side. The structural component and the processing component should be examined separately. If the available criticism or observation does not allow that distinction itself to be made with sufficient confidence, the case should not be conclusively assigned either to the COS domain or to a SCOPE-side explanation, but should instead be treated as mixed or observationally indeterminate. When multiple explanatory hypotheses remain viable, they should be stated in parallel.

SCOPE's practice of retaining multiple hypotheses must likewise not be used to erase the defeat of an individual hypothesis.

For example, if hypotheses A and B were both being retained and a new observation becomes inconsistent with hypothesis A, hypothesis A should be weakened or removed from the candidate set. The fact that hypothesis B remains viable does not mean that hypothesis A was not weakened.

If hypothesis B also becomes incompatible with subsequent observations, new hypotheses cannot be added without limit while claiming that "multiple hypotheses remain possible." Retaining multiple hypotheses is a method for dealing with uncertainty and observational indeterminacy; it is not a method for nullifying the defeat of individual hypotheses.

Therefore, the distinction between legitimate scope clarification and counterexample avoidance cannot be determined merely by asking whether the explanation already appears in existing materials. It is necessary to examine whether the distinction existed before the counterexample, whether its definition or unit of analysis has been changed to fit the counterexample, how far the distinction can be identified from observation, and whether the original proposition or individual hypothesis remains capable of being weakened, revised, or rejected. Even an existing definition must be reconsidered from the standpoint of falsifiability if it can be applied to any counterexample and has no identifiable conditions under which it fails or loses support.

Status:

"Do not treat prior existence in the theory as sufficient justification."

"Check whether the distinction existed before the counterexample appeared."

"Record distinctions introduced after a counterexample as theoretical updates."

"Do not change definitions or units of analysis to fit the current counterexample."

"When the distinction cannot be identified observationally, treat it as indeterminate."

"Retain multiple hypotheses only when multiple explanatory hypotheses actually remain viable."

"Do not treat mixed cases, observational indeterminacy, and multiple hypotheses as the same thing."

"Do not use underdeveloped concepts as conclusive escape mechanisms only when counterexamples appear."

"Treat the COS domain as a candidate explanation based on the existing distinction between cognitive structure and cognitive processing."

“When both SCOPE and COS are involved, examine the structural and processing components separately.”

“When the distinction itself cannot be identified, do not force the case into one side.”

“Do not treat a weakened or excluded individual hypothesis as unaffected merely because another explanation remains.”

“Do not use multiple-hypothesis retention to erase the defeat of an individual hypothesis.”

“Reconsider even existing concepts if they can absorb any counterexample.”

“Preserve conditions under which SCOPE or an individual proposition can lose support.”

Q100. If part of SCOPE is falsified, how much should be revised, and how much should be retained?

Related Qs: Q94–Q99, Q101–Q103

When part of SCOPE is falsified, the first task is not to decide how much of SCOPE should be revised. The first task is to identify which previously stated proposition is actually incompatible with the criticism, observation, or logical inconsistency in question.

A mere disagreement, or a claim that shows neither a logical inconsistency nor an observation incompatible with an existing proposition, is not treated as a falsification of that proposition. By contrast, when an observation or logical inconsistency is shown to be incompatible with the description or implication of an existing proposition, it is necessary to examine which proposition is affected.

However, SCOPE does not assume that internal cognitive structure itself can be directly observed. Observable output must therefore be distinguished from the interpretation or structural hypothesis inferred from that output. If an observation is incompatible with one hypothesis but remains compatible with another existing hypothesis, or if the observation does not allow the alternatives to be distinguished, the individual hypothesis may be weakened or removed from the candidate set without treating SCOPE as a whole as falsified.

Conversely, if explaining the mismatch between an observation or logical inconsistency and a prior proposition requires changing a definition, unit of analysis, boundary, layer correspondence, or similar element only after the counterexample appears, that change must not be used to erase the original counterexample. Any necessary change should be recorded as a theoretical update, while preserving the record that the pre-update proposition was incompatible with the counterexample.

The scope of revision should be examined by asking what depends on the falsified proposition. Such dependence may be used as a candidate basis for determining the affected range when it was already explicitly stated before the counterexample, or when it can be derived from existing definitions or propositions without changing them. If the dependence cannot be identified or derived from the existing materials, the scope of revision should not be fixed either narrowly or broadly.

If only one individual proposition becomes incompatible with observation or logic, and no dependence of other propositions on it can be identified, the revision may be limited to that proposition. On the other hand, if existing materials show or imply that layer definitions, connection relations, slide hypotheses, human–LLM correspondences, or other propositions depend on it, those elements must also be re-examined.

Accordingly, SCOPE should neither treat the falsification of one part as an automatic collapse of the whole nor dismiss the falsification of one part merely because the rest of the framework remains intact. The range that has been falsified, together with the range that can be shown or derived to depend on it, should be examined, and the affected parts should be revised, weakened, or rejected as necessary.

Parts that remain compatible with the counterexample and whose dependence on the falsified proposition cannot be identified or derived should not be treated as immediate revision targets solely because of that counterexample. However, changing units of analysis or boundaries after the counterexample in order to preserve those parts would not count as simple retention; it would constitute a theoretical update.

The affected range should also not be fixed in advance by assumptions such as “only peripheral propositions can be affected” or “the core must remain intact.” SCOPE does not assume that a complete dependency map already exists. If multiple central propositions become incompatible with observation or logic and cannot be accounted for through existing distinctions, the required revision may extend across a broad part of SCOPE. If observation and logic require it, the possibility that SCOPE as a whole can no longer be maintained must also remain open.

The purpose of revision is therefore not to preserve SCOPE. It is to identify which propositions have become incompatible with observation or logic, determine what dependencies can be identified or derived from the existing materials, revise, weaken, or reject the necessary range, and avoid treating the remaining parts as revision targets solely because of that counterexample.

Status:

“A mere disagreement is not treated as a falsification.”

“Distinguish observed facts from interpretations and structural hypotheses inferred from them.”

“Treat logical inconsistency as a basis for considering revision, weakening, or rejection of a proposition.”

“First identify which prior proposition is incompatible with the criticism, observation, or logic.”

“Do not force a single conclusion when the alternatives remain indistinguishable.”

“Distinguish falsification of an individual proposition from falsification of SCOPE as a whole.”

“Do not assume that a complete dependency map already exists.”

“Check whether a dependency was stated before the counterexample or can be derived from existing definitions without changing them.”

“When dependence cannot be identified or derived from existing materials, do not fix the scope of revision.”

“Re-examine propositions that can be shown or derived to depend on the falsified proposition.”

“Record post-counterexample changes to definitions, units of analysis, or boundaries as theoretical updates.”

“Do not retroactively erase the original falsification through a theoretical update.”

“Parts that remain compatible with the counterexample and whose dependence has not been established should not be revised solely because of that counterexample.”

“Do not move boundaries or units of analysis merely to preserve unaffected parts.”

“Do not assume in advance that the core must survive.”

“Preserve the possibility that SCOPE as a whole may need to be rejected.”

Q101. If a claim changes between an earlier version and the current version, how should that change be recorded?

Related Qs: Q96, Q97, Q100

In SCOPE, earlier claims should not simply be overwritten by the latest version. Changes should be preserved in a form that allows the previous claim, the reason for the change, and the revised claim to be traced.

This item does not introduce a new cognitive mechanism. It concerns an audit procedure for preserving changes in claims across versions so that those changes can later be examined.

If a definition, hypothesis, scope of application, falsification condition, or other theoretical claim presented in one version is later changed because of new observations, counterexamples, logical inconsistencies, or conceptual clarification, the revised version should state, either at the beginning of the document or in a revision history, how the claim was previously positioned, why it was changed, and how it is currently positioned.

However, not all changes should be treated as the same kind of error. Clarification of a definition, restriction of scope, updating based on new observations, and withdrawal of an individual proposition should be distinguished from one another. Typographical corrections and wording changes that do not alter meaning should also be distinguished from theoretical revisions.

Even when no explicit revision history is provided, if both the earlier and later versions remain available, it may still be possible to compare them and determine whether a claim changed. However, the reason for the change or the intent at the time of revision may not be identifiable. If the earlier version or relevant revision records are unavailable, the existence or history of the change should be treated as insufficiently verifiable.

In addition, even when a claim has been revised in the current version, the relationship between the earlier proposition and any counterexamples or logical inconsistencies recorded at that time should not be erased by the revision. These should remain part of the revision history.

In this way, SCOPE does not erase counterexamples or past inconsistencies, but preserves them as part of the record showing how the theory has changed. This item defines a method

for recording revisions; it is not a rule for determining whether SCOPE as a whole should be rejected.

Status:

- Do not simply overwrite earlier claims with the latest version.
- Preserve the previous claim, the reason for the change, and the revised claim in a traceable form.
- Distinguish different types of revision.
- Distinguish theoretical revisions from editorial corrections that do not change meaning.
- When records are incomplete, do not infer a revision history beyond what can be verified.
- Do not erase the recorded relationship between earlier claims and counterexamples or logical inconsistencies after revision.
- This item concerns revision records, not the criteria for rejecting SCOPE as a whole.

Q102. Can an individual proposition be identified as “wrong” even if SCOPE as a whole is not rejected?

Related Qs: Q94, Q96, Q97, Q98, Q99, Q100, Q101, Q103

Even if SCOPE as a whole is not rejected, an individual proposition can still be identified as wrong.

SCOPE is composed of multiple individual propositions, and the validity or invalidity of one proposition does not automatically determine the validity or invalidity of all the others.

Therefore, if an observation is obtained that is incompatible with a particular proposition, and that observation is judged to constitute a counterexample under the conditions described in Q94, that proposition may be weakened, revised, or rejected.

For example, if an observation incompatible with a particular slide relationship is established as a counterexample, that slide hypothesis becomes subject to revision or rejection. However, this does not necessarily mean that other layer definitions or connection relationships are automatically rejected as well.

At the same time, the survival of SCOPE as a whole cannot be used as a reason to erase the defeat of an individual proposition. When a counterexample is established, the dependency range of that proposition should be examined, as described in Q100, and the scope requiring revision or reconsideration should be separated accordingly.

In addition, if a new explanation or condition is introduced after a counterexample appears in order to preserve the proposition, that explanation should not be treated as though it had always been part of SCOPE. In such a case, the original proposition has not simply been preserved; rather, the theory has been updated in response to the counterexample, and this change should be recorded as such.

SCOPE therefore treats “whether the framework as a whole remains viable” and “whether an individual proposition remains viable” as separate questions. Identifying the defeat of an

individual proposition does not, by itself, mean that SCOPE as a whole must be rejected. At the same time, the continued survival of SCOPE as a whole does not cancel the defeat of that individual proposition.

Status:

“An individual proposition may be weakened, revised, or rejected even if SCOPE as a whole is not rejected.”

“The survival of SCOPE as a whole does not erase the defeat of an individual proposition.”

“The defeat of an individual proposition is not automatically converted into the rejection of SCOPE as a whole.”

“Repeated observation is not required as a necessary condition for recognizing a counterexample.”

“An explanation added after a counterexample appears is not treated as though it had been included from the beginning.”

“This item defines the distinction between partial and overall rejection; it does not define the criteria for rejecting SCOPE as a whole.”

Q103. How should SCOPE establish its own predictions or defeat conditions in the future?

Related Qs: Q94, Q98, Q99, Q100, Q101, Q102

At present, SCOPE does not have one unified prediction or defeat condition for the framework as a whole. However, some individual propositions already have incompatibility conditions or falsification conditions, while others remain unset.

Going forward, for each individual proposition under examination, SCOPE should state, as far as possible before testing begins, “what should be predicted if this proposition is correct” and “what kinds of observations or logical relationships would be incompatible with it.” It is not necessary for every possible condition to be completely specified in advance.

Here, an original prediction does not have to refer to a novel phenomenon that only SCOPE can explain. When competing models can explain the same phenomenon, the relevant task is to identify conditions under which SCOPE and those competing models predict different results, and to examine whether the models can be distinguished by observation. If no such difference can yet be specified, the evidential support for distinguishing that proposition from competing models remains unestablished. The absence of such a difference does not, by itself, mean that the proposition must immediately be declared unfalsifiable or rejected.

For example, if conditions can be specified under which a particular slide hypothesis and an independent-variable model predict different changes, those conditions can be used to test which prediction is more consistent with the results. This example illustrates how defeat conditions may be set; it does not establish the slide hypothesis itself. If a result incompatible with the SCOPE-side prediction can be identified under conditions specified in advance for that proposition, the proposition becomes subject to weakening, revision, or rejection.

Lack of identifiability is not normally a counterexample by itself. However, if a proposition claims that a state should be uniquely identifiable, then a result showing that such identification is not possible may itself become a candidate counterexample.

When setting defeat conditions, it is also important not to change the conditions after seeing the result in order to preserve the prediction. If explanations such as “it was another layer,” “the object was different,” or “it was a matter of activation level” are newly introduced only after a counterexample appears, they should not be treated as though they had been part of the original prediction. In such a case, the original proposition has not simply remained intact; the theory has been updated in response to the counterexample.

Defeat conditions do not necessarily take the same form for every proposition. For some propositions, a single incompatible result may constitute a serious counterexample. For others, evaluation may require examining a pattern across multiple results. Therefore, SCOPE should not impose one uniform threshold for the number or strength of counterexamples. Defeat conditions should be set according to the form of the proposition being tested.

When predictions and results do not match, that mismatch should not be erased. The scope of its implications should be examined only within what was stated before the counterexample appeared, or what can be derived from that material without changing it. The determination of how far the effects propagate follows Q100, while the recording of changes follows Q101.

Q103 does not introduce a new cognitive mechanism within SCOPE. It sets out an audit-oriented approach for how individual propositions should be tested in the future and under what conditions they should be treated as no longer fully sustainable. When counterexamples arise, they should not be eliminated through post hoc explanations; instead, the affected proposition and the way in which it could no longer be maintained should be preserved, with any necessary changes treated as theoretical updates.

Status:

“There is currently no single unified prediction or defeat condition for SCOPE as a whole.”

“Some individual propositions already have falsification conditions, while others remain unset.”

“For each proposition, predictions and incompatibility conditions should be specified as far as possible before testing.”

“An original prediction does not have to concern a novel phenomenon; it may instead be a prediction that distinguishes SCOPE from competing models.”

“If SCOPE cannot yet be distinguished from a competing model, evidential support for that distinction remains unestablished.”

“Lack of identifiability is not normally a counterexample, but it may become one when unique identifiability is explicitly claimed.”

“Defeat conditions should not be changed after observation in order to avoid a counterexample.”

“The form of defeat conditions depends on the proposition, and no uniform threshold for number or strength should be imposed.”

“Mismatches should not be erased; propagation follows Q100 and change records follow Q101.”

“This item defines an audit-oriented approach for setting defeat conditions and does not add a new cognitive mechanism to SCOPE.”

Q104. Why is the concept of “layer activation level” necessary separately from the slide ratio?

Related Qs: Q14, Q51–Q60, Q105–Q108

In SCOPE, slide ratios and layer activation levels are concepts used to describe different kinds of observable differences. This distinction does not mean that separate internal quantities have been confirmed to exist. Rather, it is a hypothetical distinction introduced to describe differences observed in output.

A slide ratio represents the relative distribution within each slide pair: L1–L2, L4–L5, and L6–L7. A slide ratio alone therefore does not represent the degree to which a function corresponding to each layer is estimated to be present within an observation interval. In the existing organization of SCOPE, the slide structure is likewise distinguished as representing the internal ratio within a pair rather than its activation level.

For example, even if two observations were both provisionally expressed as $L1:L2 = 50:50$, this would not necessarily mean that the degree to which the functions corresponding to those layers are estimated to appear, based on observable output, is also the same. The notation 50:50 here is only an illustrative expression and does not imply that SCOPE can currently measure slide ratios numerically. At least in the individual paper on L1, the degree of verbalization is described as varying according to the situation even within the same person, and states are organized using observational expressions such as “high,” “medium,” and “low.” These expressions are not a completed measurement scale, but a way of organizing observed differences in degree.

To distinguish this kind of difference that cannot be retained by the distribution within a pair alone, SCOPE introduces “layer activation level” as a separate hypothetical axis. The activation level referred to here is not a quantity directly measured inside the layer itself. What can be directly observed in SCOPE is the output that appears in Malkuth. Layer activation level is therefore a concept for estimating, from that output, the possible degree to which a function corresponding to a particular layer is present. Because internal cognitive structure and observable output do not correspond one-to-one, the activation level cannot be determined from a single output alone.

The reason for treating layer activation level separately from the slide ratio is therefore that some differences in degree cannot be retained by the internal distribution of a slide pair alone. However, it has not yet been established that the second axis used to represent such differences must specifically be an “activation level.” The possibility remains that the same differences could be explained by another variable or a competing model. How activation level can be estimated is addressed in Q105, how it can be distinguished observationally

from the slide ratio in Q106, competing models in Q107, and conditions that would weaken the layer-activation hypothesis in Q108.

Status: Answerable, but additional verification is required / Competing hypotheses remain

Q105. How Can Layer Activation Be Inferred from Observable Output?

Related Questions: Q6–Q14, Q88–Q93, Q104, Q106–Q108

Layer activation is not measured using a single common indicator applied in the same way to every layer. Because each layer represents a different cognitive function, the observable clues used to infer activation also differ by layer. The following are representative examples rather than an exhaustive list of inference methods or established criteria for determining activation.

For L1, possible clues may include not only the amount of speech or text, but also the extent, stability, and continuity of verbalization. However, a short utterance cannot simply be interpreted as low L1 activation, nor can a long utterance automatically be interpreted as high L1 activation. The individual paper on L1 also notes that a short utterance may represent either the result of sufficient verbalization or a case in which verbalization was insufficient. L1 therefore cannot be uniquely determined from output alone.

Short output may also arise from a communication format within COS rather than from low L1 activation. Even when cognitive content has been sufficiently verbalized, the output may remain short if it is compressed into the minimum form necessary for communication. In COS, C4 is organized as a communication format that outputs only the minimum necessary information. Short output alone therefore cannot determine the activation level of L1.

For L2, possible clues include speech, behavior, facial expression, vocal expression, and the persistence or change of responses. When emotional expression is pronounced and multiple observable output features change substantially, it may sometimes be possible to provisionally place a hypothesis of high L2 activation even from a single observation. However, no single feature, such as vocal volume or the presence of tears, can be directly equated with L2 activation. L2 is instead provisionally inferred from observable speech and behavior as an emotional configuration, while consistency with other layers is also considered.

Low activation may, in some cases, be more difficult to distinguish from a single observation. Limited speech may result from low L1 activation, but it may also reflect a person who normally communicates concisely or who has sufficiently verbalized the relevant content using only a few words. Similarly, limited emotional expression alone does not establish low L2 activation. A small amount of observable output therefore cannot itself be treated as evidence of low activation.

For L8, inference is based on how the influence of the overall cognitive structure appears in observable language, behavior, facial expression, writing, and other outputs. However, L8

itself cannot be directly observed, and no concrete criteria have yet been established for determining which output features should correspond to a particular level of L8 activation.

For L0, L3, L4, L5, L6, and L7, activation is likewise provisionally inferred from observable output corresponding to the definition of each layer. Only representative examples are presented here. For none of the layers can activation be established from a single output feature alone.

In estimating activation, the degree to which relevant features are pronounced within a single observation can be used as one source of evidence. When available, differences from previously observed output tendencies of the same subject may also provide supplementary clues. However, those usual tendencies are themselves hypotheses derived from observation rather than fixed standards. Hypotheses about such tendencies may be revised across multiple observations, but the use of multiple observations does not automatically move the analysis into the COS domain. When the process is treated as the continuing operation of a model of the subject over time, it becomes an area of connection between SCOPE and COS.

When activation is expressed as “high,” “medium,” or “low,” these labels should not be treated as a numerical scale shared across individuals. They represent relative observational levels. Therefore, “medium” activation for one subject does not necessarily represent the same amount of activation as “medium” activation for another subject. The integrated SCOPE paper also uses “high,” “medium,” and “low” as relative descriptions of slide ratios. However, those labels describe relative positions within the slide ratio. Even if the same terms are used for activation, “high,” “medium,” and “low” for slide ratio and for activation refer to different axes.

Layer activation is therefore not inferred by directly converting a single observable feature into a quantity. Instead, multiple observable clues corresponding to the definition of each layer are considered, together with their prominence within the observation, the context, the cognitive target, consistency with other layers, and, when available, differences from the subject’s previously observed output tendencies. Consistency with the broader cognitive structure does not itself establish activation. At present, no objective numerical scale or unified inference protocol applicable to all layers has been established.

Status: Answerable but requires additional verification / measurement scale not established / inference difficulty may differ across layers and activation states

Q106. How Can Layer Activation Be Distinguished from Slide Ratios at the Observational Level?

Related Qs: Q6–Q14, Q88–Q93, Q104, Q105, Q107–Q108

In SCOPE, slide ratios and layer activation may concern the same layers, but they address different questions. This is a descriptive distinction between hypotheses and does not mean that two independently existing internal quantities have been directly confirmed.

A slide ratio concerns the question, “Within a slide pair, which side is relatively more dominant?” Layer activation, by contrast, concerns the question, “How strongly is the cognitive involvement corresponding to each layer inferred to be present within the observation?”

Put simply, the slide asks “Which side is it leaning toward?”, while activation asks “How much is it operating?”

The slide relationships defined in SCOPE are L1–L2, L4–L5, and L6–L7. What is represented in each case is the relative ratio within the pair. Therefore, the slide ratio alone does not determine whether the cognitive involvement corresponding to either side is large or small. Existing SCOPE definitions also distinguish the slide model as representing a ratio rather than activation level.

By definition, within the same object and the same cognitive unit, both sides of a slide pair cannot simultaneously be assigned a high ratio. However, assigning high activation to both sides as a provisional hypothesis can be treated separately from the slide ratio.

For example, suppose two observations concerning the same object are both provisionally described for explanatory purposes as $L4 = 70$ and $L5 = 30$. This does not necessarily mean that the underlying states are identical. These numbers are illustrative only and are not established measurements.

In one observation, both the boundary judgment corresponding to L4 and the acceptance corresponding to L5 may be strongly involved in the output, while the overall relation still leans relatively toward L4. In another observation, features corresponding to both L4 and L5 may appear only weakly, while the relative relation between them remains approximately 70 to 30.

These two cases may therefore have similar slide ratios while differing in activation. In other words, the same 70:30 ratio may occur in a strongly activated state or in a weakly activated state.

Changes in activation and changes in slide ratio also do not correspond one-to-one. A change in ratio does not necessarily mean that activation on both sides changes in the same direction. Likewise, activation may change while the ratio remains relatively stable.

When provisionally distinguishing the two at the observational level, the same object and the same cognitive unit must first be held constant. The questions “Which side of the pair is relatively more dominant?” and “How strongly do the observable features corresponding to each layer appear?” should then be treated separately. This is not an established identification procedure, but a way of separating the observational questions.

However, what can actually be observed is speech, text, behavior, and other outputs appearing in Malkuth. The layer states behind those outputs remain objects of inference. Even when outputs such as keeping distance are observed, the state of L4 cannot be uniquely determined from that observation alone.

The same applies to L5. Identical speech or behavior does not necessarily imply an identical state of acceptance, and such a state cannot be uniquely determined from a single observation.

For this reason, it may sometimes be impossible, on the basis of currently available observations alone, to determine whether a change in output reflects a change in the relative ratio within the pair, a change in activation on one or both sides, or both at the same time. The distinction is not always uniquely identifiable.

When relative terms such as “high,” “medium,” and “low” are used for both ratios and activation, the same words refer to different axes. “High” or “low” in a ratio refers to relative position within the pair, whereas “high” or “low” in activation refers to the inferred degree of cognitive involvement corresponding to a given layer.

At present, no measurement scale has been established that can simultaneously quantify slide ratios and layer activation and mechanically separate them at the observational level. SCOPE therefore treats them descriptively as separate observational axes and estimates them by referring to multiple observable features, the cognitive object, context, and consistency with other layers. When the two cannot be distinguished, SCOPE does not force the observation into one interpretation.

In summary, slide ratio and layer activation are distinguished as different questions: “Which side is relatively more dominant?” and “How large is the cognitive involvement corresponding to each layer?” However, SCOPE does not yet provide a measurement method that can always uniquely identify the two. Depending on the observation, changes in ratio, changes in activation, and simultaneous changes in both may remain observationally indistinguishable.

Status: Answerable but requires further validation / Conceptually distinguishable / May be observationally indistinguishable / Measurement scale not yet established / Does not confirm internal ontological reality

Q107. Are There Competing Models That Could Explain the Same Phenomena Without Introducing Layer Activation?

Related Qs: Q6–Q14, Q51–Q55, Q88–Q93, Q104–Q106, Q108

At present, it cannot be said that any observed phenomenon has been shown to require the introduction of layer activation in order to be explained. Competing explanations that organize the same observable features without using the concept of activation may be possible.

Here, a competing model does not necessarily mean an already established theory with a recognized name in existing research. It also includes alternative descriptive methods or hypotheses that can account for the same observations as the SCOPE activation hypothesis without introducing activation as a concept. In the critical examination of SCOPE, the fact

that SCOPE can describe a phenomenon is not itself treated as evidence of validity. Competing hypotheses and alternative explanations must also be considered.

The activation hypothesis was provisionally introduced to organize an observed difference: even when the same slide state appears to be present, the magnitude of cognitive involvement may differ.

For example, even when L1–L2 appears to be in a similar slide state, there may be a difference between a case in which features corresponding to both L1 and L2 appear relatively weak and a case in which features corresponding to both appear strongly. In the latter case, one observable example may be extensive and rapid verbalization accompanied by strong emotional expression. However, the amount or speed of speech itself is not treated as identical to activation.

Likewise, in L6, two observations may both support a provisional inference that L6 is involved because some form of reasoning is present. However, there may still be a substantial difference between a case in which only minimal reasoning is expressed and a case in which multiple reasons, comparisons, conditional branches, causal relations, predictions, and structural organization are extensively developed.

Observed differences of this kind motivated the introduction of the activation hypothesis.

However, the existence of such observational differences does not mean that they can only be explained through a common activation axis. The fact that the activation hypothesis arose from observation and the claim that activation is an irreplaceable explanation are separate issues.

At least two types of competing approaches can be considered.

The first is to describe observed features individually without combining them into a cross-layer construct called “activation.” In the case of L6, for example, the range of reasoning, comparisons, conditional branches, causal relations, predictions, and structural organization could be recorded separately without summarizing them as a single value or level of “L6 activation.”

The second is to allow the magnitude of involvement to be provisionally described separately for each layer without treating those magnitudes as instances of a single activation axis shared across all layers. This competing approach does not deny the description of magnitude itself. Rather, it questions the assumption that such magnitudes must be organized along one common observational axis across all layers.

Similarly, features specific to each layer could be recorded independently—for example, the amount and stability of verbalization in L1 or observable features associated with emotional expression in L2—without combining them into one common construct called activation. The observed differences could still be described in this way.

Some output differences may also be explainable through slide ratios, differences in the cognitive object, or contextual differences. However, these cannot be used as universal

substitutes for activation. As discussed in Q106, slide ratios concern which side of a pair is relatively more dominant. They do not easily express a situation in which the ratio remains similar while the magnitude of cognitive involvement corresponding to each layer differs.

Some output differences may also be explained through transmission formats described in COS. However, COS cannot be treated as a general substitute for activation. COS organizes cognitive processing, whereas SCOPE is a coordinate system for inferring cognitive structure. COS therefore cannot be used automatically to fill undefined elements on the SCOPE side.

The current activation hypothesis therefore has an observational motivation. However, that fact alone does not establish either the necessity or the validity of the activation hypothesis.

If a model that retains individual observable features, or a model that describes magnitude separately for each layer, can explain the same phenomena with fewer assumptions, and if introducing activation provides no additional discrimination, prediction, or comparability, then the need to treat activation as a common observational axis would be weakened.

Conversely, if introducing activation makes it possible to organize states consistently that are difficult to distinguish using slide ratios or lists of individual features alone, and if those differences can be shown to remain reproducible across observations, the rationale for retaining activation would become stronger.

At present, no comparative validation has been conducted to determine whether a model using activation, a model describing only individual observable features, or a model describing magnitude separately for each layer provides the best discrimination, reproducibility, or comparability with the fewest assumptions.

For this reason, activation is currently positioned within SCOPE as a descriptive hypothesis for organizing observed differences in the magnitude of cognitive involvement along a common axis. However, it has not yet been established that this organization is necessary or that it performs better than descriptions based on individual features or layer-specific magnitudes.

This question also does not determine whether existing models in cognitive science, psychology, or AI research directly compete with the SCOPE activation hypothesis, because no external theoretical comparison has been conducted here. Comparison with external theories should be treated as a separate stage from the internal examination of SCOPE.

In summary, the activation hypothesis has an observational motivation, but it has not been shown that the observed differences can only be explained through activation. Competing models capable of describing the same observable phenomena without introducing activation may exist. At present, activation has not been established as an indispensable explanatory axis, and its necessity must be tested through comparison with competing models.

Status: Competing hypotheses exist / Observational motivation for introduction exists / Indispensability of activation not confirmed / Necessity as a common axis not validated /

Comparative validation not conducted / Comparison with external existing models not conducted

Q108. What Observations Would Weaken the Hypothesis of Layer Activation?

Related Qs: Q6–Q14, Q51–Q55, Q88–Q93, Q104–Q107

Layer activation is not treated as a directly observable internal state. It is a hypothesis used to infer, from outputs appearing in Malkuth, the magnitude of cognitive involvement corresponding to each layer. Therefore, the fact that activation itself cannot be directly observed does not by itself weaken the hypothesis. In SCOPE, the internal layers themselves are not directly observable and are instead inferred hypothetically from observable outputs.

The activation hypothesis would be weakened if the observational differences it is intended to explain could also be consistently explained by models that do not use activation, while introducing activation failed to provide additional discriminability, reproducibility, or predictive value.

For example, suppose two cases appear to have similar slide states but are inferred to differ in the magnitude of cognitive involvement. If those differences can repeatedly be described sufficiently using only individual observable features—such as amount of speech, extent of reasoning, emotional expression, context, cognitive object, or transmission format—the need to introduce activation as an additional concept would be weakened.

The hypothesis would also be weakened if cases inferred to involve “high activation” and “low activation” were observed repeatedly but no stable observable differences emerged between them, or if the inferred activation level varied substantially depending on the observation period or evaluator. In particular, if different evaluators repeatedly inferred substantially different activation levels from the same observational material and those differences could not be explained consistently, the validity of activation as a reproducible observational construct would decrease.

The rationale for maintaining activation as an independent hypothesis would also weaken if a model using activation were compared with a model describing only individual observable features and the activation-based model provided no additional benefit in distinguishing new states, organizing observations, making predictions, or enabling comparison.

There is also a separate weakening condition for activation as a shared concept. Even if it remains useful to describe differences in the magnitude of involvement within each layer, the hypothesis of organizing those differences under a common concept of “activation” would be weakened if the inference criteria differed substantially across layers and grouping them under one common concept produced no additional discrimination or comparability. In that case, the description that “each layer can vary in magnitude” might remain, while the need to organize those differences under a shared activation construct would weaken.

As discussed in Q106, changes in slide ratio and changes in activation cannot always be distinguished observationally. Observational indistinguishability by itself does not mean that the activation hypothesis is false. However, if this indistinguishability persists and no stable difference can be demonstrated between models that include activation and models that do not, support for treating activation as a necessary shared concept would weaken. This would not constitute an immediate rejection of the activation hypothesis. Rather, it would mean that observational support for its necessity has not been obtained.

Another serious problem would arise if every counterexample could be absorbed after the fact by introducing activation, another layer, a different cognitive object, slide ratio, COS, L8, or some other concept, thereby allowing the activation hypothesis to survive regardless of the observation. A hypothesis for which no possible counterexample can weaken it loses falsifiability. In the critical examination of SCOPE, this kind of unlimited recovery through additional concepts is treated as a risk of “ghostification.”

However, the fact that a single case can be explained without activation would not by itself invalidate the activation hypothesis as a whole. If activation fails to distinguish states within a particular layer or under a particular condition, this may weaken only its scope of application or its inference criteria. In contrast, if across multiple layers and different observational conditions the introduction of activation repeatedly fails to provide greater discriminability, reproducibility, or predictive value than models without activation, the rationale for treating activation as a hypothesis shared across SCOPE would be substantially weakened. Partial weakening and weakening of the overall hypothesis should therefore be distinguished.

Accordingly, the activation hypothesis is not weakened merely because an alternative explanation is possible. Its justification would weaken if competing models without activation consistently explained the same observational differences and repeated comparison showed that adding activation provided no new discriminability, reproducibility, prediction, or comparability.

At present, such comparative validation has not been conducted. It therefore remains unknown whether the activation hypothesis meets any of these weakening conditions. Unresolved issues should remain explicitly unresolved, and when competing interpretations cannot be distinguished observationally, that indistinguishability itself should be recorded.

Status: Weakening conditions can be specified / Falsifiability considered / Comparative validation against competing models not conducted / Reproducibility not validated / Necessity as a shared concept not confirmed / Partial and overall weakening distinguished